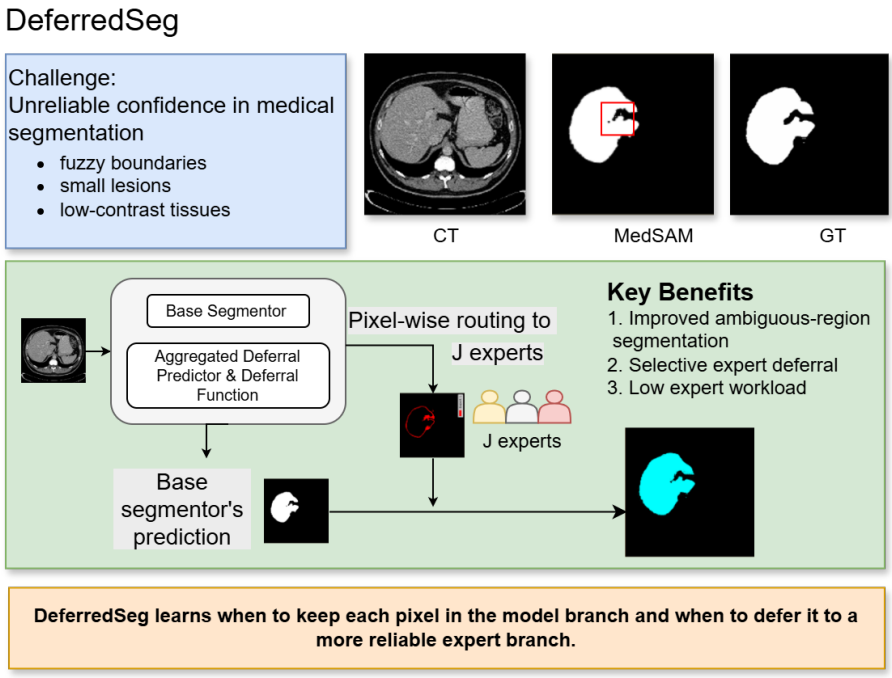
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Graphical Abstract



Highlights

- *DeferredSeg*, a pixel-wise deferral framework for medical image segmentation.
- A Deferral Collaboration loss learns pixel-wise routing between the model and expert branches.
- An aggregated deferral predictor encourages spatially coherent deferral regions.
- A load-balancing penalty mitigates routing collapse in multi-expert settings.
- Experiments with synthetic expert masks and offline clinician annotations evaluate collaborative segmentation.

DeferredSeg: A Multi-Expert Deferral Framework for Medical Image Segmentation

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Abstract

Segmentation models based on deep neural networks demonstrate strong generalization for medical image segmentation. However, they often exhibit overconfidence or underconfidence, leading to unreliable confidence scores for segmentation masks, especially in ambiguous regions. This limits the reliability of fully automatic segmentation in ambiguous regions. Motivated by the learning-to-defer (L2D) paradigm, we introduce *DeferredSeg*, a pixel-wise Human–AI collaborative deferral framework that learns whether to retain model predictions or defer difficult pixels or regions to expert branches.

DeferredSeg extends the base segmentor with an aggregated deferral predictor and additional routing channels that dynamically route each pixel to either the base segmentor or human expert. To train this routing efficiently, we introduce a pixel-wise surrogate collaboration loss that supervises deferral decisions. In addition, we introduce a spatial-coherence loss that aligns the aggregated expert-assignment probability with the learned routing decisions, encouraging locally consistent deferral patterns.

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Beyond single-expert deferral, we further extend the framework to a multi-expert setting by introducing multiple experts with different reliability profiles for collaborative decision-making. To mitigate overloading or underutilization of individual experts, we further design a load-balancing penalty that discourages routing collapse and promotes more balanced utilization across expert branches. We evaluate *DeferredSeg* on multiple medical segmentation benchmarks using MedSAM and CENet as the base segmentors, and further compare it with an expert-assisted patch-correction baseline. Experimental results show that *DeferredSeg* consistently improves System-level segmentation performance while selectively deferring difficult pixels or regions to more reliable expert branches. Moreover, the proposed framework is backbone-compatible and can be instantiated with different segmentation architectures that provide intermediate feature representations.

Keywords: Learn to defer, Human–AI collaboration, Medical image segmentation

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1. Introduction

Medical image segmentation is a fundamental task in medical data analysis, aiming to assign a semantic label to every pixel or voxel in an image. Accurate delineation of anatomical structures or pathological regions is crucial for a broad range of downstream applications, including computer-aided diagnosis, treatment planning, and intraoperative navigation [1]. Compared to natural image segmentation, medical segmentation is inherently more challenging due to heterogeneous imaging modalities, low-contrast boundaries, irregular shapes, and the scarcity of high-quality annotations. Traditional approaches have demonstrated effectiveness, but deep neural networks now dominate medical image segmentation. Skip-connection architectures, notably U-Net [2] and its variants [3, 4], remain strong clinical baselines. Recent Transformer-based designs, including kMaXU [5] , further enhance performance through global-context and multi-scale representation learning.

Although these models have demonstrated remarkable generalization ability, recent studies report that these deep models often produce *overconfident* or *underconfident* predictions [6]. This tendency is particularly evident in ambiguous or low-contrast regions, such as fuzzy boundaries, small lesions, and distribution shifts [7]. In high-stakes medical segmentation, such issues pose a critical risk: incorrect predictions delivered with high certainty may mislead clinicians and cause severe consequences.

Therefore, relying solely on automated outputs is unsafe; incorporating *Human–AI collaboration* that integrates clinical expertise into the decision-making process for high-risk cases is essential. Several interactive segmentation methods have been proposed, such as MIDeepSeg [8] and TIS [9], which reduce the amount of user interaction required. However, these methods still require the user to make decisions on the entire image, rather than focusing on smaller, more ambiguous regions.

The learning-to-defer paradigm addresses this issue by allowing a model to either make its own prediction or defer to an expert when confidence is low [10, 11]. Unlike conventional uncertainty estimation, L2D treats referral as an explicit decision and routes unreliable predictions to experts [12].

However, existing L2D methods operate at the instance level, making them unsuitable for dense prediction tasks where uncertainty varies spatially at the pixel level. In medical segmentation, such instance-wise deferral cannot resolve localized ambiguities, including unclear tissue boundaries and heterogeneous lesion textures. This motivates a region-selective collaboration strategy for dense prediction, where only ambiguous regions are referred for expert review instead of requiring full-image correction.

In this paper, we propose *DeferredSeg*, a pixel-wise learning-to-defer framework for dense medical image segmentation. The core of our framework is a pixel-wise deferral mechanism designed to support Human–AI collaboration through a deferral predictor and a learned routing function. These components dynamically route

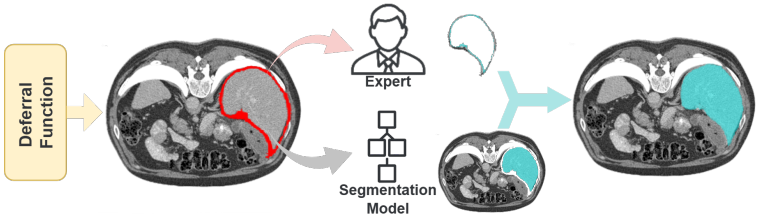


Figure 1: Overview of *DeferredSeg*. A deferral predictor produces a pixel-wise deferral map that triages image regions between human-expert branches and the base segmentation model, enabling spatially adaptive Human–AI collaboration for deferral.

pixel-wise predictions, determining whether to accept the base segmentation model's prediction or to defer ambiguous regions to human experts, as illustrated in Fig. 1. In the intended deployment, experts provide local corrections only for the pixels or regions selected by the learned deferral function.

Our main contributions are:

1. **A Novel Deferral-Aware Segmentation Framework.** We introduce *DeferredSeg*, a pixel-wise deferral-aware framework for Human–AI collaborative medical image segmentation. It features a dynamic deferral predictor that intelligently routes pixel-wise decisions to either the base model or expert branches. This is enabled by two tailored loss functions: a surrogate deferral collaboration loss for efficient training and a spatial-coherence loss to ensure deferred regions are spatially coherent and reliable.
2. **A Multi-Expert Setting for Balanced Collaboration.** Our method introduces multiple experts to facilitate collaborative decision-making. This includes a load-balancing penalty that mitigates routing collapse and discourages severe overload or underutilization of individual expert branches.
3. **Backbone-compatible Instantiation.** We instantiate *DeferredSeg* with MedSAM and further validate its transferability on CENet, showing that the proposed routing mechanism can be attached to different segmentation backbones.
4. **Comprehensive Evaluation.** Experiments with synthetic expert masks and

pre-existing clinician annotations show that *DeferredSeg* improves System-level segmentation performance while quantifying pixel-level expert-routing ratios.

2. Related Work

2.1. Medical Image Segmentation

2.1.1. Classical and Deep Learning Methods

Early medical image segmentation relied on handcrafted methods such as energy-based models, threshold-based methods [13], and graph-based methods [14]. Although effective for high-contrast structures, these methods are sensitive to noise, parameter tuning, and anatomical variability, and have gradually been replaced by data-driven learning approaches [15].

Building on the limitations of classical methods, learning-based approaches, especially CNN-based architectures, have become dominant in medical image segmentation. Representative models include U-Net [2], DeepMedic [4], and the self-configuring nnU-Net framework [16], while variants such as Attention U-Net [3] improved boundary delineation. More recently, Transformer-based models such as UNETR [17] further improve long-range dependency modeling. Hybrid CNN–Transformer architectures have also been developed to combine local texture modeling with global context modeling, such as CTC-Net [18] and TBConvL-Net [19]. Despite these advances, robust segmentation remains challenging for low-contrast tissues, irregular boundaries, and rare anatomical structures [1].

2.1.2. Foundation Models for Image Segmentation

Foundation models have recently attracted increasing attention in image segmentation due to their strong generalization capabilities and potential for zero-shot or few-shot transfer across diverse modalities and anatomical structures. Among them, the Segment Anything Model (SAM) [20] enables flexible prompt-based mask generation using spatial prompts such as points, bounding boxes, and prior masks. Inspired

by SAM's success in natural image segmentation, several works have adapted this paradigm to the medical domain. MedSAM [21] fine-tunes SAM on a large corpus of medical images while preserving its prompt-based architecture, establishing a strong baseline for generalizable medical segmentation. MA-SAM [22] improves spine segmentation through multi-atlas-guided pseudo-mask prompts, while SegMIC [23] uses in-context learning to support generalizable medical image segmentation. Despite these advances, prompt-based foundation models still face challenges in clinically critical scenarios involving low-contrast regions, irregular boundaries, small lesions, and ambiguous structures.

2.1.3. Human–AI-assisted and Multi-expert Segmentation

Human–AI-assisted segmentation improves reliability by incorporating user interaction, expert review, or annotation feedback into the segmentation process. Interactive segmentation methods refine masks using clicks, scribbles, bounding boxes, or other prompts. Representative methods such as MIDeepSeg [8] and TIS [9] show that explicit interaction can improve segmentation quality, but they usually require users to inspect the image and provide interaction signals during inference, which may still be time-consuming for scattered errors or ambiguous structures.

Uncertainty-based referral limits expert review to unreliable or informative samples or regions. Uncertainty-aware segmentation architectures, such as UCTNet [24], further exploit uncertainty to guide feature extraction or model computation in medical image segmentation. Uncertainty-based methods often rely on confidence scores, entropy, prediction variance, or model disagreement to identify unreliable predictions [25].

Patch-based error correction provides a local form of expert intervention. For example, PatchCorr [26] selects uncertain patches and corrects them with expert input, avoiding full-image manual correction. However, such methods usually operate at a coarse patch level and depend on uncertainty-guided selection, without explicitly

learning pixel-wise routing among the model and multiple expert branches. Moreover, the correction ratio or expert-intervention budget must be manually specified in advance, which may require task-specific tuning.

Expert specialization has also been explored in MoE-style medical segmentation. Recent methods such as MoE-SAM [27], topology-preserving MoE segmentation [28], and medical language MoE [29] demonstrate the value of specialized computational experts in medical image analysis. These methods demonstrate the value of specialized experts, but their experts are typically computational modules inside the network rather than external expert branches used for inference-time deferral.

Overall, existing studies have advanced interactive correction, uncertainty-guided review, active annotation, patch-level quality control, and computational expert specialization. However, most of them rely on predefined interaction or uncertainty-selection strategies, or use experts as internal model modules. Therefore, they do not directly address how to automatically decide, at the pixel level, which regions should remain with the segmentation model and which difficult regions should be assigned to external human expert branches. This gap motivates a pixel-wise deferral formulation for medical image segmentation.

2.2. Learning to Defer

L2D integrates algorithmic or human expertise into automated decision systems by learning whether to make a prediction or defer the input to one or more experts. It extends the classic *learning to reject* paradigm [30] by replacing simple rejection with expert-aware triage. Madras et al. [31] formulated learning to defer to human experts, while De et al. [32] further extended deferral to regression.

Direct optimization of the discontinuous deferral objective is intractable, so L2D research has focused on designing consistent, differentiable surrogates. A cost-sensitive, consistent surrogate loss using a softmax-based formulation was first proposed in [10]. A calibrated one-vs-all objective was introduced in [33]. For multi-expert scenarios,

a consistent and calibrated surrogate was developed in [34], and a two-stage approach with a matching consistent loss was proposed in [35].

L2D has also been extended beyond classification, including regression with expert deferral [36], sequential healthcare decision-making [37] and semi-supervised learning with sparse expert availability [38]. These studies demonstrate the generality of L2D for Human–AI collaboration across tasks.

However, most L2D methods make instance-level routing decisions, which are unsuitable for dense segmentation where uncertainty varies spatially and errors concentrate along fine anatomical boundaries (e.g., fuzzy borders, low-contrast tissues, small lesions). This motivates our pixel-wise deferral formulation for medical image segmentation.

3. Preliminaries

We begin by formalizing the *general* learning-to-defer problem with multiple experts, and then provide the formulation of the true risk for the 0–1 loss and its surrogates for optimization.

3.1. Data and Model

Suppose there are J experts involved in L2D tasks. For a K-class classification problem, each training example in L2D is represented as a triplet $(\mathbf{x}, y, \mathbf{m}) \sim \mathcal{D}$, where $\mathbf{x} \in \mathcal{X}$ is the input, $y \in \{1, \ldots, K\}$ is the ground-truth label, and $\mathbf{m} \in \mathbb{R}^J$ is the vector of expert's prediction with element $\mathbf{m}_i \in \{1, \ldots, K\}$. In addition to a classifier $h : \mathcal{X} \to \mathbb{R}$, a deferral function $r : \mathcal{X} \to \{0, 1, \ldots, J\}$ determines whether to defer and, if so, to which expert. Specifically, when $r(\mathbf{x}) = 0$, the classifier makes the prediction; when $r(\mathbf{x}) = j$, the classifier abstains and defers the decision to the j-th expert.

3.2. The True Risk of the 0–1 Loss

The true risk $\mathcal{L}$ of L2D is extended from a generalization of learning with rejection, as proposed in [39], which can be formalized as a 0–1 loss as follows:

$$\mathcal{L}_{0-1}(h, r) = \mathop{\mathbb{E}}_{\mathbf{x},y,\mathbf{m}}\Big[\mathop{\mathbb{I}}_{r(\mathbf{x})=0} \mathop{\mathbb{I}}_{h(\mathbf{x})\neq y} + \sum_{j=1}^{J} \mathop{\mathbb{I}}_{r(\mathbf{x})=j} \mathop{\mathbb{I}}_{\mathbf{m}_j\neq y} \Big]. \tag{1}$$

Here, $\mathbb{I}[\cdot]$ denotes the indicator function, which equals 1 if the condition inside the brackets is satisfied and 0 otherwise.

The estimated classifier and deferral function pair $(\hat{h}, \hat{r})$ can be obtained by minimizing Eq. (1) on the training set $\mathcal{D}$. However, $\mathcal{L}_{0\text{-}1}$ is discontinuous and non-convex, making direct minimization impractical; therefore, we adopt differentiable and consistent surrogates for optimization.

3.3. The Softmax-Based (SM) Surrogate Loss

The softmax-based loss is commonly used in L2D. We augment the label space as $\mathcal{Y}^{\perp} = \mathcal{Y} \cup \{\perp_1, \ldots, \perp_J\}$, where $\perp_j$ represents the option to defer to the j-th expert. We build a predictor $f : \mathcal{X} \to \mathbb{R}^{K+J}$, composed of the classifier h and deferral function r. f can also be decomposed into $K + J$ functions: $f_k : \mathcal{X} \to \mathbb{R}$ for $k \in \{1, \ldots, K\}$, and $f_{\perp,j} : \mathcal{X} \to \mathbb{R}$ for $j \in \{1, \ldots, J\}$, where k, j denote the class and expert indexes, respectively. These $K + J$ functions are combined using a softmax-parameterized surrogate loss:

$$\begin{aligned}\mathcal{L}^{J}_{SM}(f_1, \ldots, f_K, f_{\perp,1}, \ldots, f_{\perp,J}; \mathbf{x}, y, \mathbf{m}) = \\ - \sum_{j=1}^{J} \mathbb{I}[\mathbf{m}_j = y] \log\left(\frac{\exp f_{\perp,j}(\mathbf{x})}{\sum_{y'\in\mathcal{Y}^{\perp}} \exp f_{y'}(\mathbf{x})}\right) - \log\left(\frac{\exp f_y(\mathbf{x})}{\sum_{y'\in\mathcal{Y}^{\perp}} \exp f_{y'}(\mathbf{x})}\right).\end{aligned} \tag{2}$$

The first term maximizes the deferral function $f_{\perp,j}$ if the j-th expert's prediction is correct. The second term ensures that the function f_k is maximized for the true label.

At test time the classifier prediction is obtained by selecting the largest score among the first K outputs, and deferral is decided as follows:

$$r(\mathbf{x}) = \begin{cases} 0, & \text{if } f_h(\mathbf{x}) > \max_{j\in\{1,\dots,J\}} f_{\perp,j}(\mathbf{x}), \\ \arg\max_{j\in\{1,\dots,J\}} f_{\perp,j}(\mathbf{x}), & \text{otherwise.} \end{cases} \tag{3}$$

4. Method

Our methodology extends learning to defer (L2D) from instance-level prediction to pixel-wise medical image segmentation. This section presents the overall framework of *DeferredSeg*, the pixel-wise routing formulation, the training objectives, and the MedSAM-based instantiation.

4.1. Overall Framework

DeferredSeg is a backbone-compatible pixel-wise deferral framework with multi-expert collaboration, designed to work with a base segmentation model that provides dense feature representations and segmentation predictions. We augment the model's output channels by adding multiple routing channels, where one corresponds to the base model and the remaining correspond to specialized expert branches. These routing channels provide the architectural basis for pixel-wise deferral by producing branch-specific routing logits across spatial regions. An auxiliary CNN-based aggregated deferral predictor enforces spatially coherent routing, and training combines collaboration, spatial-coherence, and load-balancing losses.

Given an input image, the base segmentor first produces dense features and segmentation predictions. The routing function then assigns each pixel either to the model branch or to an expert branch. The final System output retains the base prediction for model-routed pixels and uses the selected expert prediction for deferred pixels.

4.2. Pixel-Wise L2D with Multiple Experts

4.2.1. Output Channel Extension and Pixel-wise Routing

Let $\mathbf{x} \in \mathcal{X}$ be an image of size $H \times W$, and let $i \in \{1, \ldots, HW\}$ index pixels; $\mathbf{x}^i$ denotes the value at the i-th pixel. The ground-truth label of $\mathbf{x}^i$ is $y^i \in \{0, 1\}$ for a binary segmentation task. To enable pixel-wise L2D with J experts, we extend the base segmentor by adding J+1 routing channels. The original segmentation output channels remain unchanged. Therefore, the label space is defined $\mathcal{Y}^{\perp} = \mathcal{Y} \cup \{\perp_0, \perp_1, \cdots, \perp_J\}$, where the labels from $\perp_0$ to $\perp_J$ correspond to the additional routing channels. Unlike the classification setting in Sec. 3, in our segmentation setting, the predictor is composed of the base segmentor $f_{\text{seg}} : \mathcal{X} \to \mathbb{R}^{H \times W}$ and the deferral function $f_{\perp} : \mathcal{X} \to \mathbb{R}^{(J+1) \times H \times W}$. In practice, the base segmentor produces a probability prediction $\hat{y}^i \in [0, 1]$ for the foreground class at each pixel $\mathbf{x}^i$. The deferral function outputs per-pixel logits $f_{\perp,j}(\mathbf{x}^i)$ with $j \in \{0, \ldots, J\}$. A larger $f_{\perp,0}(\mathbf{x}^i)$ indicates a stronger preference to route pixel i to channel 0 (the base segmentor), while a larger $f_{\perp,j}(\mathbf{x}^i)$ $(j > 0)$ indicates a stronger preference to route it to channel j (the j-th expert). For the j-th expert branch, $\mathbf{m}^i_j \in \{0, 1\}$ denotes the j-th expert's binarized prediction at pixel $\mathbf{x}^i$.

4.2.2. The Deferral Collaboration Loss

To enable pixel-wise L2D, we adapt the softmax surrogate introduced in Sec. 3 (Eq. (2)) and generalize it as the *Deferral Collaboration (DC) Loss* for dense segmentation, which is defined as:

$$
\begin{aligned}
\mathcal{L}_{\text{DC}}(f_{seg}, f_{\perp,0}, \ldots, f_{\perp,J}; \mathbf{x}^i, y^i, \mathbf{m}^i) = \frac{1}{HW} \sum_{i=1}^{HW} \Bigg[& - \sum_{j=1}^{J} \mathbb{I}[\mathbf{m}^i_j = y^i] \log \left(\frac{\exp f_{\perp,j}(\mathbf{x}^i)}{\sum_{t=0}^{J} \exp f_{\perp,t}(\mathbf{x}^i)} \right) \\
& - \sum_{j=1}^{J} \Big(1 - \mathbb{I}[\mathbf{m}^i_j = y^i]\Big) \log \left(1 - \frac{\exp f_{\perp,j}(\mathbf{x}^i)}{\sum_{t=0}^{J} \exp f_{\perp,t}(\mathbf{x}^i)} \right) \\
& - \mathbb{I}[\tilde{y}^i = y^i] \log \left(\frac{\exp f_{\perp,0}(\mathbf{x}^i)}{\sum_{t=0}^{J} \exp f_{\perp,t}(\mathbf{x}^i)} \right) - \Big(y^i \log \hat{y}^i + (1 - y^i) \log (1 - \hat{y}^i) \Big) \Bigg].
\end{aligned} \tag{4}
$$

where $\mathbb{I}[\cdot]$ denotes the indicator function, $\tilde{y}^i = \mathbb{I}[\hat{y}^i \geq 0.5]$ denotes binarized segmentor prediction. Specifically:

- The first term increases the routing probability assigned to expert j when its prediction matches the ground truth. The second term suppresses the routing probability assigned to expert j when its prediction is incorrect.
- The third term maximizes the segmentor branch deferral score $f_{\perp,0}(\mathbf{x}^i)$ if the segmentor prediction $\tilde{y}^i$ matches the ground truth.
- The fourth term is the pixel-wise binary cross-entropy (BCE) on the segmentor output $\hat{y}^i$, which supervises the foreground prediction itself.

The loss encourages each pixel to select the branch with the more reliable prediction. The thresholded prediction $\tilde{y}^i$ is used only to construct routing supervision, whereas the segmentation branch itself is optimized through the soft probability $\hat{y}^i$ in the BCE term.

During inference, the routing probability map $\mathcal{M}$ of the j-th expert at pixel i is obtained by applying a softmax over all routing logits:

$$\mathcal{M}_j^i = \frac{\exp(f_{\perp,j}(\mathbf{x}^i))}{\sum_{j=0}^{J} \exp(f_{\perp,j}(\mathbf{x}^i))}. \tag{5}$$

4.2.3. Spatial-Coherence Loss

To generate a smooth deferral map and ensure that deferred regions are spatially coherent and reliable, we introduce an additional *aggregated deferral predictor* built with two convolutional layers and enforce consistency between its predictions and those of the deferral function score $f_{\perp,j}(\cdot)$ using a *spatial coherence (SC) loss*.

Motivation: *State-of-the-art medical image segmentation models often employ Vision Transformer (ViT)-based decoders. However, these architectures can be less effective for data-efficient deferral learning on small datasets, as ViTs exhibit weak*

inductive bias for spatial relevance due to their emphasis on long-range interactions [40]. To address this, we introduce a CNN-based predictor to complement the ViT by providing intra-patch locality, enabling the capture of fine structures within patches and improving the spatial consistency of expert-assignment regions.

Let $\tilde{\mathcal{M}} \in [0,1]^{H\times W}$ denote the aggregated expert-assignment probability map predicted by the aggregated deferral predictor. Given the per-pixel routing probabilities $\mathcal{M}_j^i$, each pixel of the binary supervision mask $\bar{\mathcal{M}}$ is computed as

$$\bar{\mathcal{M}}^i = \begin{cases} 0, & \text{if} \quad \arg\max_{j\in\{0,\dots,J\}} \mathcal{M}_j^i = 0 \\ 1, & \text{otherwise.} \end{cases} \tag{6}$$

Here, $\bar{\mathcal{M}}$ is not a ground-truth segmentation mask; instead, it serves as a pseudo supervision signal indicating whether a pixel is routed to the model branch or to any expert branch according to the current routing logits. The role of the SC loss is therefore not to recover semantic labels, but to encourage spatial consistency in the deferral pattern. The goal of the *spatial coherence (SC) loss* is to minimize the discrepancy between $\tilde{\mathcal{M}}$ and $\bar{\mathcal{M}}$:

$$\mathcal{L}_{\text{SC}} = \frac{1}{HW}\sum_{i=1}^{HW}\Big[\underbrace{-\Big(\bar{\mathcal{M}}^i \log \tilde{\mathcal{M}}^i + (1-\bar{\mathcal{M}}^i)\log(1-\tilde{\mathcal{M}}^i)\Big)}_{\text{BCE}} + \beta_1 \underbrace{\left(\tilde{\mathcal{M}}^i - \sum_{j=1}^{J}\mathcal{M}_j^i\right)^2}_{\text{MSE}} + \beta_2\, \tilde{\mathcal{M}}^i\Big]. \tag{7}$$

We adopt two commonly used losses with different properties. The first term is the pixel-wise binary cross-entropy (BCE). The second term is the mean squared error (MSE) aligning $\tilde{\mathcal{M}}^i$ with the sum of all expert routing probabilities $\mathcal{M}_j^i$. Additionally, the last term penalizes excessive deferral and discourages the trivial all-defer solution. β_1 and β_2 are two weighting hyperparameters.

4.2.4. Load-Balancing Penalty

In multi-expert settings, routing collapse may occur if the deferral function over-relies on a subset of experts, leading to underuse or overload. To prevent this and encourage fair utilization, we introduce a *Load-Balancing (LB) Penalty* that regulates the average pixel-wise routing allocation of each expert branch in a batch of samples. For each expert branch $j \in \{1, \ldots, J\}$, let the average routing ratio be

$$\rho_j = \frac{1}{BHW} \sum_{b=1}^{B} \sum_{i=1}^{HW} \mathcal{M}_j^{(b,i)}, \tag{8}$$

where B is the batch size, and $\mathcal{M}_j^{(b,i)}$ is the routing probability of expert j for pixel i in the b-th image.

We impose lower and upper bounds $\mathbf{l} = [l_1, \ldots, l_J]$ and $\mathbf{u} = [u_1, \ldots, u_J]$ on the routing ratios, which are treated as configurable hyperparameters to avoid expert underuse or overload. The penalty is defined as:

$$\mathcal{L}_{\text{LB}} = \sum_{j=1}^{J} \Big[\max(\rho_j - u_j, 0) + \max(l_j - \rho_j, 0) \Big]. \tag{9}$$

This penalty discourages extreme concentration or vanishing probability mass in the soft routing distribution and helps stabilize multi-expert training.

4.2.5. The Learning Objective

In the multi-expert setting, the training objective is:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{DC}} + \lambda_1 \, \mathcal{L}_{\text{SC}} + \lambda_2 \, \mathcal{L}_{\text{LB}}, \tag{10}$$

where λ_1 and λ_2 are scalar weights that control the relative importance of spatial coherence and load balancing. For the setting of one expert, we can omit the $\mathcal{L}_{\text{LB}}$.

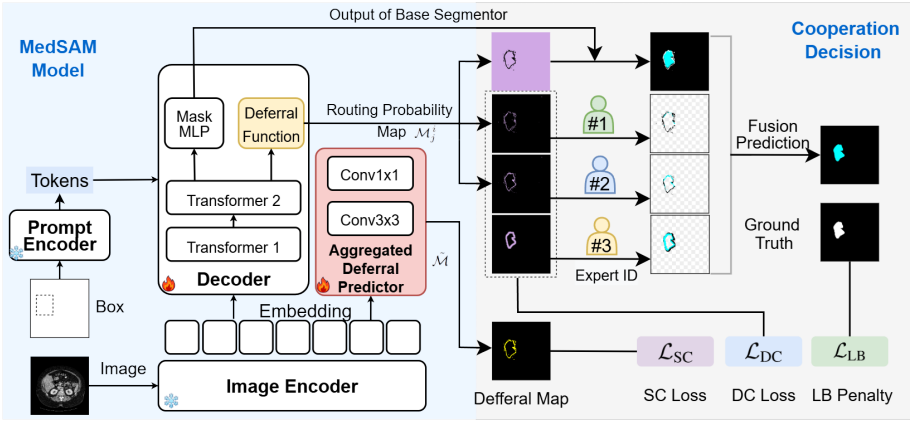


Figure 2: Overall pipeline of the proposed *DeferredSeg* framework, illustrated with a MedSAM-based instantiation. The deferral function $f_\perp$ denotes the routing head that outputs per-pixel routing logits. The routing probability maps $\mathcal{M}_j$ denote softmax-normalized branch-assignment probabilities. The binary deferral mask $\tilde{\mathcal{M}}$ denotes the hard model/expert routing decision derived from the routing probability maps. The aggregated deferral predictor denotes the auxiliary CNN-based predictor that outputs $\tilde{\mathcal{M}}$, a single-channel aggregated expert-assignment probability map used for spatial-coherence supervision.

4.2.6. Inference-time System Fusion

After training, the final System prediction is obtained by pixel-wise routing and fusion. For each pixel i, the hard routing decision is determined by the largest routing logit:

$$r^i = \arg\max_{j\in\{0,\dots,J\}} f_{\perp,j}(\mathbf{x}^i), \tag{11}$$

where $r^i = 0$ indicates that pixel i is assigned to the model branch, and $r^i > 0$ indicates that it is assigned to one of the expert branches. The final System prediction is then obtained as

$$\hat{y}^i_{\text{sys}} = \begin{cases} \tilde{y}^i, & r^i = 0, \\ \mathbf{m}^i_{r^i}, & r^i \in \{1,\dots,J\}, \end{cases} \tag{12}$$

where $\tilde{y}^i = \mathbb{I}[\hat{y}^i \geq 0.5]$ is the binarized prediction of the base segmentor, and $\mathbf{m}^i_{r^i}$ is the prediction from the selected expert branch. Therefore, the System prediction uses synthetic expert masks or clinician annotations only for pixels routed to expert branches, while pixels routed to the model branch are predicted by the base segmentor.

4.3. MedSAM-based Instantiation

DeferredSeg is designed as a backbone-compatible deferral framework that can be integrated with segmentation models providing dense feature representations and pixel-wise predictions. In this work, we instantiate *DeferredSeg* with MedSAM to leverage its strong cross-domain generalization for medical images, as illustrated in Fig. 2. Concretely, we keep the MedSAM encoder intact and extend the decoder by adding a deferral function and an aggregated deferral predictor, enabling pixel-wise routing to the base or expert branches without altering MedSAM's native mask outputs. This integration preserves compatibility with MedSAM while adding minimal overhead to support multi-expert collaboration.

MedSAM Encoder: The encoder of MedSAM remains frozen during training, ensuring that the model retains its cross-domain generalization ability. This frozen encoder processes input images to generate high-level feature representations that are then passed to the modified decoder.

Modified Decoder: To enable routing without altering MedSAM's native outputs, we augment its mask decoder with an additional deferral function. Concretely, MedSAM uses three learnable mask tokens and the same number of MLPs to produce native mask logits. We extend this design by adding $(J{+}1)$ *routing tokens*: the first corresponds to the segmentor branch and the remaining J correspond to the expert branches ($j{=}1, \ldots, J$). In parallel, we expand the number of MLPs from size 3 to $3 + (J{+}1)$, where the first three MLPs are initialized from MedSAM's pretrained weights and fine-tuned during training, whereas the newly added $(J{+}1)$ heads are *randomly initialized* to produce per-pixel routing logits.

Aggregated Deferral Predictor: To integrate it into MedSAM, we extend the mask decoder with an additional CNN-based aggregated deferral predictor. Concretely, given the decoder features $U \in \mathbb{R}^{C\times H\times W}$, the predictor applies a 3×3 convolution, ReLU activation, and a 1×1 convolution to produce a single-channel aggregated expert-assignment probability map, $\tilde{\mathcal{M}} \in [0,1]^{H\times W}$. This predictor runs in

parallel with the existing routing logits $\{f_{\perp,j}\}_{j=0}^{J}$. While the routing logits are used for softmax-based deferral, the CNN predictor provides an explicit and spatially smooth estimate of the aggregated expert-assignment probability. During training, both outputs are supervised jointly by the spatial-coherence loss in Eq. (7). At inference time, the aggregated deferral predictor is used only for visualizing the overall deferral tendency and does not participate in the hard routing decision or final System fusion.

Through this MedSAM-based instantiation, *DeferredSeg* leverages strong pretrained medical image representations while adding pixel-wise deferral capability for uncertain or challenging regions. Importantly, the same routing formulation can also be attached to other segmentation backbones, as further supported by the CENet transfer experiment.

5. Experiments

We conduct extensive experiments to evaluate *DeferredSeg* from four perspectives: comparison with strong baselines, performance across organs and modalities, scalability to multiple experts, and ablation of key components.

5.1. Datasets & Metrics

We evaluate the proposed method under two expert settings, as summarized in Tab. 1: (1) *synthetic-expert benchmarks*, including PROMISE12, LiTS, and AMOS22, where expert annotations are simulated to study controllable collaboration patterns; and (2) *offline clinician-annotation benchmarks*, including Chaksu, QUBIQ, and CURVAS, where pre-existing annotations from individual clinicians are used to instantiate expert branches.

We report DSC, Jaccard, and Sensitivity in all experiments.

5.2. Implementation Details

5.2.1. Preprocessing

For the MedSAM- and CENet-based experiments, 2D slices are resized to 1024×1024 and 224×224, respectively. CT intensities are clipped to $[-250, 250]$ HU and normalized to $[0, 1]$, while MRI volumes are normalized using per-volume min–max scaling. Random flips and 90° rotations are applied during training.

5.2.2. Expert Sources

The expert branch in *DeferredSeg* conceptually represents the local correction provided by a human expert after the model defers selected pixels or regions. In the intended deployment, the learned deferral function first identifies these regions, and experts then provide corrections only within their assigned regions. In our experiments, synthetic expert masks and pre-existing clinician annotations serve as proxies for these prospective expert corrections. The former enables controlled analysis of region-dependent expert reliability, while the latter preserves natural inter-annotator variability. In both settings, the masks provide branch predictions only for pixels selected by the learned routing function; they are not used as routing inputs and do not determine where deferral occurs. These settings allow us to evaluate the routing behavior and System performance of *DeferredSeg* when expert outputs are available.

Synthetic expert masks. To reduce reliance on costly manual annotations or external experts, we propose the `SynthExpertSeg` function to instantiate J synthetic experts. Each expert $j \in \{1, \ldots, J\}$ is defined by an accuracy triplet $([\mathrm{FG}_j,\ \mathrm{BG}_j,\ \mathrm{BD}_j])$, specifying its performance on foreground (FG), background (BG), and boundary (BD) regions. These synthetic experts are intentionally designed with imperfect and region-dependent reliability profiles, rather than as oracle experts. Therefore, the System cannot obtain a perfect mask simply by deferring more pixels; instead, it must learn whether the model branch or a given expert branch is more reliable for each pixel,

Table 1: Summary of the datasets.

Dataset	Modality	#Volumes/Images	ROI
PROMISE12 [41]	MRI	80	Prostate
LiTS [42]	CT	201	Liver / Tumor
AMOS22 [43]	CT / MRI	600	15 organs
Chaksu [44]	Fundus	1345 images	Optic Disc / Cup
QUBIQ	MRI	39 / 55 images	Brain / Prostate
CURVAS	CT	90 volumes	Kidney / Liver

Table 2: Region-specific accuracy profiles (%) used to generate the synthetic expert masks in the comparative experiments.

# Expert	FG	BG	BD
1	92	98	98
2	85	99	94
3	75	95	90

which is the core setting where learning-to-defer is meaningful. The standard expert types and their accuracy settings used in our comparative experiments (SubSec. 5.3) are summarised in Tab. 2.

Real clinician annotations. For all offline clinician-annotation benchmarks, E1 is fixed as the single expert branch in the one-expert setting, while E1, E2, and E3 are used simultaneously in the three-expert setting. STAPLE (Simultaneous Truth and Performance Level Estimation) is a probabilistic annotation-fusion method that estimates a reference consensus mask from multiple clinician annotations while accounting for their estimated annotation reliability.

For Chaksu, the released STAPLE consensus mask is used as the reference mask, while the annotations from Expert 1, Expert 2, and Expert 3 are used as E1, E2, and E3, respectively. For QUBIQ Brain Growth and QUBIQ Prostate, the first three individual annotations are used as E1, E2, and E3, while the remaining annotations are combined using STAPLE to construct the reference masks. For CURVAS, the first three clinician annotations are used as E1, E2, and E3, and their STAPLE consensus is used as the reference mask. Therefore, the annotations used to construct the QUBIQ reference masks are disjoint from those used as expert branches. In contrast, the Chaksu and CURVAS reference masks are consensus estimates derived from the same annotation pool used to instantiate the expert branches.

5.2.3. Hyperparameter Configuration

The overall training objective is defined in Eq. (10), with loss weights set to $\lambda_1 = 1.0$ and $\lambda_2 = 5.0$. Components are configured as: **SC Loss:** $\beta_1 = 0.5$ and $\beta_2 = 0.1$; **LB Penalty:** Upper and lower routing-allocation bounds for J=3 experts are $\mathbf{u} = [0.35, 0.25, 0.30]$, $\mathbf{l} = [0.05, 0.05, 0.05]$. Training details are provided in Appendix Appendix C.

5.2.4. Hardware and Software Environment

All experiments are conducted using Python 3.10, PyTorch 2.1, and MONAI 1.3 on NVIDIA RTX 4090 GPUs, with a single GPU used unless otherwise specified.

5.3. Comparative Results

5.3.1. Quantitative Results

We evaluate *DeferredSeg* against strong automated baselines, including nnU-Net v2 [45], CENet [46], zero-shot MedSAM, and supervised fine-tuned MedSAM. These fully automatic baselines provide model-only reference performance, allowing us to quantify the potential benefit of pixel-wise deferral when most pixels are still handled by the model branch and only difficult pixels are deferred to expert branches. To make this additional expert cost explicit, we report the Model-only branch, the deferral ratio, and the expert routing allocation distribution in Tab. 5, showing that *DeferredSeg* defers only a small fraction of pixels to expert branches.

We further include PatchCorr [26] as an expert-assisted patch-correction baseline. PatchCorr selects uncertain local patches according to the prediction-margin uncertainty of MedSAM and replaces the predictions within these patches using the same expert source as the one-expert setting of *DeferredSeg*. For a fair comparison of expert intervention, the correction budget of PatchCorr is matched to the one-expert deferral ratio of *DeferredSeg* on each dataset.

Synthetic-expert evaluation.

Table 3: Comparison across multiple segmentation tasks under different evaluation and training settings (%). The methods are grouped into zero-shot inference, full task-specific training, expert-assisted correction, and fine-tuned MedSAM-based comparison. MedSAM-FT (Fine-tuned) serves as the direct backbone reference for DeferredSeg. Gray values following ↑ denote absolute percentage-point improvements over MedSAM-FT. Row-wise best values are shown in **bold**.

Dataset	Metric	Zero-shot	Full task-specific training		Expert-assisted correction	Compare to the Fine-tuned		
		MedSAM ZS	nnU-Net v2	CENet	PatchCorr	MedSAM FT	DeferredSeg 1E	DeferredSeg 3E
Prostate (PROMISE12)	DSC	84.16	90.49	68.96	92.24	86.17	**97.06** ↑10.89	96.00 ↑9.83
	Jaccard	73.01	82.70	54.62	85.98	76.08	**94.38** ↑18.30	92.46 ↑16.38
	Sens.	76.70	91.78	75.20	94.45	84.97	95.33 ↑10.36	**96.42** ↑11.45
Liver (LiTS)	DSC	91.64	96.62	93.61	92.86	91.69	**97.96** ↑6.27	96.15 ↑4.46
	Jaccard	86.43	93.48	89.52	88.83	86.49	**96.30** ↑9.81	93.70 ↑7.21
	Sens.	93.98	**98.37**	93.85	98.08	94.06	96.83 ↑2.77	95.47 ↑1.41
Tumor (LiTS)	DSC	59.62	68.04	75.35	65.42	59.20	**91.01** ↑31.81	81.82 ↑22.62
	Jaccard	47.83	52.72	64.91	54.91	47.55	**85.81** ↑38.26	74.59 ↑27.04
	Sens.	76.18	84.99	76.38	87.75	75.97	**88.99** ↑13.02	84.39 ↑8.42
Spleen	DSC	89.21	94.29	89.62	93.80	94.57	97.75 ↑3.18	**98.14** ↑3.57
	Jaccard	81.74	90.83	84.79	89.33	90.77	95.86 ↑5.09	**96.59** ↑5.82
	Sens.	90.78	95.21	89.41	95.41	95.02	98.06 ↑3.04	**98.34** ↑3.32
Right Kidney	DSC	89.39	90.26	91.34	93.76	94.83	**98.34** ↑3.51	**98.34** ↑3.51
	Jaccard	81.98	85.99	86.47	89.05	90.85	**96.85** ↑6.00	96.84 ↑5.99
	Sens.	91.70	85.86	91.27	96.46	95.25	**98.69** ↑3.44	98.49 ↑3.24
Left Kidney	DSC	89.63	93.67	91.61	93.42	94.74	98.14 ↑3.40	**98.16** ↑3.42
	Jaccard	82.33	88.99	86.92	88.63	90.86	96.60 ↑5.74	**96.63** ↑5.77
	Sens.	93.11	93.21	91.39	97.62	94.67	**98.72** ↑4.05	98.25 ↑3.58
Gallbladder	DSC	79.60	76.73	62.58	90.82	86.69	**96.57** ↑9.88	95.27 ↑8.58
	Jaccard	68.58	68.31	45.66	84.62	78.77	**93.88** ↑15.11	91.82 ↑13.05
	Sens.	79.56	81.95	50.48	92.20	88.46	**97.27** ↑8.81	96.62 ↑8.16
Esophagus	DSC	65.72	82.71	66.44	90.96	79.67	96.14 ↑16.47	**96.44** ↑16.77
	Jaccard	51.50	72.07	55.24	84.10	68.37	92.78 ↑24.41	**93.28** ↑24.91
	Sens.	67.36	84.43	65.63	88.46	78.58	**96.50** ↑17.92	96.43 ↑17.85
Liver (AMOS22)	DSC	86.78	97.17	90.90	89.29	95.15	97.54 ↑2.39	**97.87** ↑2.72
	Jaccard	78.67	94.57	86.80	82.85	91.82	95.48 ↑3.66	**96.07** ↑4.25
	Sens.	91.25	97.37	90.64	95.99	95.44	98.44 ↑3.00	**98.70** ↑3.26
Stomach	DSC	81.56	87.27	77.07	89.81	90.69	**96.70** ↑6.01	96.67 ↑5.98
	Jaccard	70.82	80.14	70.37	83.32	84.98	**94.03** ↑9.05	93.98 ↑9.00
	Sens.	81.04	89.78	76.35	92.84	90.46	**97.62** ↑7.16	**97.62** ↑7.16
Aorta	DSC	86.45	94.85	86.35	93.57	94.61	98.00 ↑3.39	**98.20** ↑3.59
	Jaccard	78.02	90.40	80.95	89.90	90.36	96.34 ↑5.98	**96.68** ↑6.32
	Sens.	87.59	93.75	85.29	94.69	94.35	98.11 ↑3.76	**98.47** ↑4.12
Inferior vena cava	DSC	77.59	87.84	75.42	92.37	85.42	**97.02** ↑11.60	97.01 ↑11.59
	Jaccard	65.58	78.84	66.16	86.43	76.61	**94.39** ↑17.78	94.38 ↑17.77
	Sens.	75.48	88.42	85.20	89.20	83.88	97.17 ↑13.29	**97.23** ↑13.35
Pancreas	DSC	69.30	82.06	62.46	80.51	83.90	**95.68** ↑11.78	95.39 ↑11.49
	Jaccard	56.06	71.11	47.18	71.06	74.11	**92.19** ↑18.08	91.67 ↑17.56
	Sens.	77.23	85.44	56.61	90.46	86.92	**96.54** ↑9.62	95.82 ↑8.90
Duodenum	DSC	62.14	74.75	38.54	75.44	79.83	94.63 ↑14.80	**94.86** ↑15.03
	Jaccard	48.34	62.07	25.74	65.21	68.50	90.38 ↑21.88	**90.72** ↑22.22
	Sens.	69.37	74.59	30.89	82.02	81.53	**95.40** ↑13.87	95.38 ↑13.85
Bladder	DSC	81.64	87.11	75.38	89.50	89.17	**95.99** ↑6.82	**95.99** ↑6.82
	Jaccard	71.80	80.74	61.78	83.42	82.72	93.27 ↑10.55	**93.29** ↑10.57
	Sens.	81.94	94.39	69.68	93.21	92.45	**97.68** ↑5.23	97.62 ↑5.17
Prostate/ Uterus	DSC	81.11	85.40	41.64	93.16	88.73	97.09 ↑8.36	**97.19** ↑8.46
	Jaccard	69.65	75.83	29.57	87.77	80.77	94.58 ↑13.81	**94.78** ↑14.01
	Sens.	76.65	88.46	36.72	92.42	91.44	97.21 ↑5.77	**97.26** ↑5.82

The main quantitative comparison is summarized in Tab. 3, while the complete results with standard deviations and the Expert/Model branches are provided in Appendix D. For *DeferredSeg*, we report three perspectives: *System* performance, *Expert-only* performance, and *Model-only* performance, following the inference-time deferral rule defined in Eq. (12). The computational overhead of introducing the deferral mechanism is also evaluated in Appendix M, showing only a small increase in model complexity and inference cost over the base MedSAM.

Across PROMISE12, LiTS, and AMOS22, *DeferredSeg* achieves improved System-level performance over the automated baselines in most settings. PROMISE12 evaluates a single mid-sized organ with ambiguous gland boundaries, where both the one- and three-expert configurations improve overlap and sensitivity by selectively routing difficult boundary pixels. Additional leave-one-center-out validation on PROMISE12 further confirms that *DeferredSeg*, particularly with three experts, improves overlap-based performance under center shifts with limited expert deferral (Appendix Appendix F). LiTS contains large homogeneous liver regions together with small, low-contrast tumors; *DeferredSeg* improves both tasks, with particularly clear gains for tumor segmentation, while the three-expert setting remains competitive for liver segmentation. AMOS22 further evaluates 13 abdominal organs with diverse sizes and contrasts. On this multi-organ benchmark, *DeferredSeg* performs strongly against nnU-Net v2, CENet, and MedSAM, with statistically significant improvements over MedSAM for most organs (Tab. 7), especially for small, low-contrast, and anatomically complex structures.

It should be noted that the expert branches are not assumed to be globally superior to the model branch. As shown in the complete branch-level results, in several cases such as Left kidney, Spleen, and Inferior vena cava, the Expert-only branch is even weaker than the Model-only branch. This indicates that the gain of *DeferredSeg* does not come from simply replacing the model prediction with stronger expert predictions. Instead, *DeferredSeg* learns to selectively defer pixels or regions to expert branches

only when they are locally more reliable, while retaining the model prediction in regions where the model branch is more competent.

Real-expert evaluation.

In this subsection, the expert branches are instantiated using clinician annotations. The main quantitative comparison is summarized in Tab. 4, while the complete results with standard deviations and the Expert/Model branches are provided in Appendix E.

Table 4: Real-expert evaluation across Chaksu, QUBIQ, and CURVAS (%). MedSAM-FT serves as the direct backbone reference for DeferredSeg. Gray values following the arrows denote absolute percentage-point changes relative to MedSAM-FT. Row-wise best values are shown in **bold**.

Dataset	Metric	Zero-shot	Task-specific training	Expert-assisted correction	Compare to the Fine-tuned		
		MedSAM ZS	CENet	PatchCorr	MedSAM FT	DeferredSeg 1E	DeferredSeg 3E
Chaksu	DSC	94.97	97.20	90.37	96.76	**97.36** ↑0.60	97.17 ↑0.41
	Jaccard	90.68	94.64	82.78	93.89	**94.87** ↑0.98	94.60 ↑0.71
	Sens.	91.51	96.67	84.11	96.69	96.30 ↓0.39	**96.74** ↑0.05
Brain Growth (QUBIQ)	DSC	66.98	86.76	82.10	83.15	86.68 ↑3.53	**89.33** ↑6.18
	Jaccard	50.57	79.10	72.37	71.28	78.52 ↑7.24	**81.51** ↑10.23
	Sens.	80.86	86.04	84.38	85.03	86.49 ↑1.46	**88.20** ↑3.17
Prostate (QUBIQ)	DSC	93.39	94.23	95.62	94.77	**96.73** ↑1.96	95.71 ↑0.94
	Jaccard	89.24	89.36	91.68	90.16	**94.56** ↑4.40	92.17 ↑2.01
	Sens.	92.16	**96.72**	96.29	96.20	93.69 ↓2.51	92.76 ↓3.44
Kidney (CURVAS)	DSC	67.94	93.17	91.32	90.70	94.48 ↑3.78	**95.80** ↑5.10
	Jaccard	54.93	88.78	86.75	84.41	90.00 ↑5.59	**91.98** ↑7.57
	Sens.	88.90	92.45	89.49	91.83	92.67 ↑0.84	**93.97** ↑2.14
Liver (CURVAS)	DSC	92.51	93.84	92.39	93.14	93.55 ↑0.41	**95.80** ↑2.66
	Jaccard	87.74	88.43	87.24	88.39	89.93 ↑1.54	**91.94** ↑3.55
	Sens.	92.68	93.65	92.16	94.17	**94.88** ↑0.71	94.04 ↓0.13

Across the three clinician-annotation benchmarks, the benefit of multi-expert deferral depends on task-specific expert complementarity. For Chaksu, where both the model and clinician-annotation branches are already strong, the one-expert configuration achieves the best overlap-based performance, while the multi-expert setting remains competitive and outperforms PatchCorr. On QUBIQ, the benefit of multiple experts is task-dependent: they improve brain-growth segmentation, whereas the one-expert setting performs best for prostate segmentation. Notably, for prostate segmentation, *DeferredSeg* still benefits from selectively using locally reliable expert regions even when the Expert-only branch is weaker than the model branch. On CURVAS, multi-expert collaboration is particularly beneficial for kidney segmentation, whereas

liver segmentation presents a metric-dependent trade-off between the one- and three-expert settings. These results show that *DeferredSeg* can exploit locally reliable expert regions, but that the value of adding experts depends on their complementarity and the target task.

Expert routing allocation and deferral ratio.

Tab. 5 reports the routing distribution. Across both synthetic- and real-expert benchmarks, the model branch processes the large majority of pixels, while only a limited subset is deferred to expert branches. The multi-expert setting generally increases the total deferral ratio, particularly on PROMISE12, QUBIQ-Prostate, and the CURVAS tasks, but most pixels remain assigned to the model branch. The individual expert routing allocations are not necessarily uniform because the routing function selects expert branches according to their local pixel-level reliability. These results show that the System improvements are achieved through selective pixel-wise deferral rather than full-image expert intervention.

Table 5: Expert routing allocation and deferral ratios under the one- and three-expert settings for synthetic expert masks and clinician annotations. Values are reported as percentages. Deferred denotes the percentage of pixels routed to expert branches, and Model denotes the percentage retained by the model branch.

Expert source	Dataset	One expert			Three experts		
		Deferred	E1	Model	Deferred	E1 / E2 / E3	Model
Synthetic expert masks	PROMISE12	4.32	4.32	95.68	10.28	0.80 / 8.32 / 1.16	89.72
	LiTS-Liver	1.57	1.57	98.43	1.45	1.45 / < 0.01 / < 0.01	98.55
	LiTS-Tumor	0.48	0.48	99.52	0.46	0.40 / 0.04 / 0.02	99.54
	Spleen	0.55	0.55	99.45	0.72	0.70 / 0.01 / 0.01	99.28
	Right Kidney	0.48	0.48	99.52	0.66	0.62 / 0.03 / 0.01	99.34
	Left Kidney	0.53	0.53	99.47	0.67	0.62 / 0.04 / 0.01	99.33
	Gallbladder	0.38	0.38	99.62	2.57	0.59 / 1.98 / 0.00	97.43
	Esophagus	0.11	0.11	99.89	0.19	0.11 / 0.07 / 0.01	99.81
	Liver	1.75	1.75	98.25	3.02	2.01 / 1.00 / < 0.01	96.98
	Stomach	1.29	1.29	98.71	2.23	1.32 / 0.88 / 0.02	97.77
	Aorta	0.19	0.19	99.81	1.67	0.59 / 1.00 / 0.08	98.33
	Inferior vena cava	0.17	0.17	99.83	0.25	0.18 / 0.07 / 0.00	99.75
	Pancreas	0.72	0.72	99.28	0.43	0.28 / 0.11 / 0.04	99.57
	Duodenum	0.45	0.45	99.55	0.50	0.40 / 0.07 / 0.02	99.50
	Bladder	1.04	1.04	98.96	1.03	0.89 / 0.13 / 0.01	98.97
	Prostate/Uterus	0.61	0.61	99.39	0.49	0.49 / 0.00 / 0.00	99.51
Clinician annotations	Chaksu	1.98	1.98	98.02	3.11	1.02 / 1.11 / 0.98	96.89
	QUBIQ-Brain Growth	1.27	1.27	98.73	3.37	1.41 / 0.97 / 0.98	96.63
	QUBIQ-Prostate	1.28	1.28	98.72	7.88	2.97 / 1.91 / 3.00	92.12
	CURVAS-Kidney	3.22	3.22	96.78	5.94	1.41 / 2.53 / 2.00	94.06
	CURVAS-Liver	1.63	1.63	98.37	5.32	1.69 / 2.27 / 1.36	94.68

Transferability to fully supervised CENet (Tab. 6).

Table 6: CENet baseline versus CENet with L2D under the three-expert setting (mean ± standard deviation, %). Row-wise best values are **bold**.

Dataset	Metric	CENet	Ours (3 experts)		
			System	Expert	Model
Prostate (PROMISE12)	DSC	68.96 ± 0.16	**88.85 ± 0.07**	89.80 ± 0.82	71.98 ± 6.69
	Jaccard	54.62 ± 0.13	**80.47 ± 0.03**	82.01 ± 1.22	59.89 ± 3.87
	Sens.	75.20 ± 0.10	**92.79 ± 2.38**	94.24 ± 0.87	72.41 ± 7.32
Liver (LiTS)	DSC	93.61 ± 0.10	**95.16 ± 0.48**	94.19 ± 1.00	94.75 ± 1.86
	Jaccard	89.52 ± 0.07	**91.73 ± 0.71**	89.04 ± 0.97	91.95 ± 2.70
	Sens.	93.85 ± 0.14	**94.03 ± 0.71**	95.67 ± 0.71	95.57 ± 1.44
Tumor (LiTS)	DSC	75.35 ± 0.52	**79.06 ± 0.14**	84.42 ± 0.22	69.49 ± 0.31
	Jaccard	64.91 ± 0.36	**65.37 ± 0.17**	76.08 ± 0.25	53.24 ± 0.29
	Sens.	76.38 ± 0.65	**78.57 ± 0.11**	79.03 ± 0.19	59.62 ± 0.27
Spleen	DSC	89.62 ± 0.25	**93.94 ± 0.09**	90.41 ± 0.05	89.52 ± 2.46
	Jaccard	84.79 ± 0.26	**88.58 ± 0.16**	82.55 ± 0.08	85.56 ± 1.20
	Sens.	89.41 ± 0.28	**90.89 ± 0.01**	85.96 ± 0.10	88.60 ± 3.11
Right Kidney	DSC	91.34 ± 0.86	**95.35 ± 0.12**	88.31 ± 0.05	91.67 ± 0.15
	Jaccard	86.47 ± 0.93	**91.11 ± 0.08**	79.26 ± 0.08	87.65 ± 0.08
	Sens.	91.27 ± 1.16	**94.00 ± 0.06**	81.20 ± 0.01	91.31 ± 0.97
Chaksu	DSC	97.20 ± 0.06	**97.27 ± 0.17**	97.32 ± 0.15	97.19 ± 0.21
	Jaccard	94.64 ± 0.08	**94.74 ± 0.30**	94.84 ± 0.27	94.60 ± 0.39
	Sens.	**96.67 ± 0.15**	95.87 ± 0.52	95.94 ± 0.51	95.82 ± 0.52

To verify backbone compatibility, we integrate *DeferredSeg* into CENet under the same three-expert setting. Although CENet already provides a strong baseline, adding *DeferredSeg* brings further gains across several synthetic-expert benchmarks and remains competitive on the real-expert Chaksu benchmark. These results indicate that the proposed deferral mechanism is not specific to MedSAM and can generalize to different segmentation backbones.

Statistical significance analysis.

We conduct one-sided paired Wilcoxon signed-rank tests on case-level DSC scores. The MedSAM-based one- and three-expert configurations are compared with MedSAM-FT and PatchCorr, while the CENet-based configuration is compared with CENet-FT under the same backbone. When multiple random seeds are available, per-case DSC scores are first averaged across seeds. For a baseline b, the paired difference is defined as $d_i^{(b)} = \mathrm{DSC}_{DeferredSeg,i} - \mathrm{DSC}_{b,i}$. The null hypothesis assumes no positive shift in the paired differences, whereas the alternative hypothesis assumes a positive shift, indicating that *DeferredSeg* achieves higher case-level DSC than the corresponding

Table 7: Summary of statistically significant improvements in case-level DSC. Values denote the number of significant comparisons over the total number of comparisons. Detailed results are provided in Appendix H.

Baseline	1E	3E
MedSAM-FT	15/16	15/16
PatchCorr	16/16	16/16
CENet-FT	–	6/6

Table 8: Analysis of each loss component through ablation. We report the overall System performance in the main text; detailed Expert and Model results are provided in the appendix Table I.16.

J	Loss	DSC	Jaccard	Sens.
1	DC	96.86±0.92	94.00±1.73	96.80±1.89
	+ SC	**97.17±0.63**	**94.57±1.17**	**97.29±1.24**
3	DC	95.39±1.64	91.51±2.95	97.13±2.23
	+ SC	95.79±1.71	92.14±3.07	97.18±2.31
	+ LB	**96.59±1.46**	93.52±2.69	97.16±2.19
	+ SC, LB	96.58±1.04	**93.56±1.87**	**98.34±0.47**

baseline.

As summarized in Tab. 7, *DeferredSeg* achieves significant improvements over MedSAM-FT in 30 of 32 comparisons and over PatchCorr in all 32 comparisons. Under the CENet backbone, *DeferredSeg* achieves statistically significant improvements over CENet-FT on all six evaluated segmentation tasks.

5.3.2. Qualitative Results

We present per-slice comparisons and deferral heatmaps to visualize System predictions and expert-assignment probabilities.

Expert Routing and System Predictions. We compare per-slice results for the one- and three-expert configurations against MedSAM, nnU-Net v2, and CENet. As shown in Fig. 3(a), *DeferredSeg* produces sharper contours and fewer false positives by deferring ambiguous pixels, particularly near organ boundaries and in small, low-contrast regions. In the one-expert setting, routing focuses on the most difficult pixels, whereas in the multi-expert setting it remains boundary-concentrated but becomes more fine-grained, allowing different experts to handle different types of difficult boundary pixels. Thus, deferral is selective rather than uniformly distributed across the image. Additional qualitative results are provided in Appendix Appendix G.

Deferral Heatmap Visualization. To inspect where deferral tends to occur, we visualize the *deferral heatmap* together with the ground-truth mask in Fig. 3(b). The heatmap represents the continuous aggregated expert-assignment probability $\tilde{M}$, with

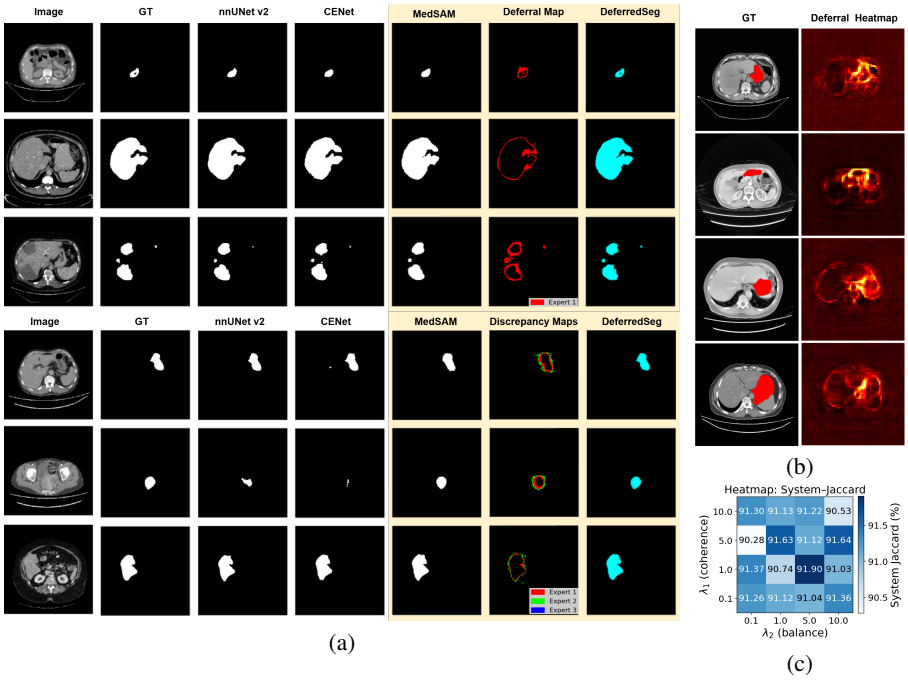


Figure 3: (a) Qualitative results under the *1-expert* and *3-expert* settings. (b) Deferral heatmap, where brighter regions indicate higher expert-assignment probability. (c) System Jaccard heatmap over coherence weight λ_1 and balance weight λ_2, with the best performance obtained at λ_1=1 and λ_2=5.

brighter regions indicating a stronger tendency to defer pixels to expert branches, typically around ambiguous boundaries or low-contrast regions. In contrast, the final routed expert pixels are obtained by applying the argmax rule in Eq. (11) to the routing logits.

5.4. Ablation Study and Analysis

5.4.1. Selection of Hyperparameters (λ_1, λ_2)

A grid search is conducted on PROMISE12 over $\lambda_1, \lambda_2 \in \{0.1, 1, 5, 10\}$, resulting in 16 configurations. For brevity, the main text visualises only the System performance heatmap based on the Jaccard index (Fig. 3(c)). The complete numerical results covering all three branches (System / Expert / Model) and metrics (Jaccard / DSC / Sensitivity) are reported in Appendix J (Tab. J.17), together with the full set of nine heatmaps (Fig. J.7). Across all configurations, the performance consistently peaks at λ_1=1 and

Table 9: System performance with increasing numbers of complementary experts on PROMISE12. Values are averaged over three random seeds.

J	Experts	Jaccard	DSC	Sens.
1	E1	91.04	95.24	95.54
2	E1, E6	89.03	94.10	93.58
3	E1, E2, E3	92.45	95.99	96.38
4	E1, E2, E3, E4	94.06	96.83	97.43
5	E1, E2, E3, E4, E5	**94.40**	**96.88**	**97.46**

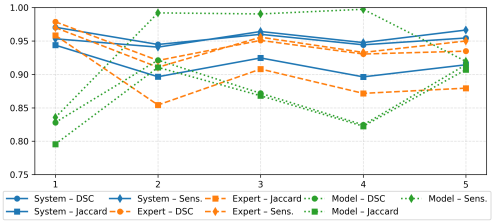


Figure 4: Expert scalability. System DSC and Jaccard peak at J=1, dip at J=2, and level off for $J \geq 3$. Sensitivity peaks at J=5.

λ_2=5, which are therefore adopted in the main experiments. To further assess whether this hyper-parameter choice is dataset-specific, we additionally conduct a sensitivity analysis on LiTS-Liver in Appendix J.1.

5.4.2. Effect of Loss Components

We conduct ablation experiments on PROMISE12 to examine the contribution of each loss component under both single-expert (J=1) and multi-expert (J=3) settings. All ablation configurations are independently trained with three random seeds. Results are summarised in Tab. 8.

Single-expert. The results show that adding the SC Loss consistently improves the overall segmentation performance, indicating that explicit supervision on the deferral map remains beneficial in the single-expert setting.

Multi-expert. With three experts and only the DC Loss, the system obtains 95.39±1.64% DSC, which is lower than the single-expert case, indicating that simply increasing the number of experts can dilute routing. Adding the SC Loss brings small but consistent improvements, while the LB Penalty yields a larger gain. Combining SC and LB produces the best overall results, confirming that spatial coherence and balanced expert utilization are both important for effective multi-expert collaboration.

5.4.3. Expert Scalability Analysis

To assess scalability, we evaluate *DeferredSeg* on PROMISE12 with $J \in \{1, 2, 3, 4, 5\}$ synthetic experts under identical training settings. The expert sets are drawn from a fixed pool with comparable FG/BG accuracy (see Appendix K, Tab. K.19), so performance differences primarily reflect how well the deferral mechanism fuses multiple experts.

From the quantitative results in Appendix K (Tab. K.20), the System obtains the highest overlap with a single expert (J=1). As the number of experts increases, DSC/Jaccard generally decrease or fluctuate, while Sensitivity improves and peaks at J=5.

Overall, Fig. 4 and Tab. K.20 jointly reveal a clear precision–recall trade-off: adding more experts improves Sensitivity but slightly reduces overlap due to increased routing dispersion. Nevertheless, the System maintains strong DSC/Jaccard even at J=5, indicating that our pixel-wise deferral mechanism scales robustly with the number of experts on the MedSAM backbone.

Complementary Experts. We further study scalability under *complementary expert* configurations. Seven synthetic experts with distinct accuracy profiles on FG, BG, and BD regions are defined (Appendix L, Tab. L.21); by selecting different subsets of these experts, we emulate varying levels of complementarity and specialisation.

As shown in Tab. 9, the benefit of multiple experts is not strictly monotonic. The two-expert setting performs worse than the single-expert setting, suggesting that heterogeneous experts may introduce routing ambiguity when their strengths are not sufficiently complementary. However, with three or more complementary experts, System performance recovers and further improves, reaching the best Jaccard, DSC, and Sensitivity at J=5. These results suggest that multi-expert deferral benefits from complementary regional strengths rather than simply from including more experts. The full System / Expert / Model results with standard deviations are reported in Ap-

pendix L (Tab. L.22).

6. Conclusion

We propose *DeferredSeg*, a pixel-wise deferral framework for medical image segmentation. By combining routing channels with deferral-aware, spatial-coherence, and load-balancing objectives, *DeferredSeg* provides a pixel-wise mechanism for Human–AI collaboration between a base segmentor and one or more expert branches. Experiments with synthetic expert masks and clinician annotations show that *DeferredSeg* improves System-level segmentation performance across multiple datasets, particularly in small, low-contrast, and ambiguous regions.

The main strengths of *DeferredSeg* are its pixel-wise deferral mechanism, its support for multi-expert collaboration, and its compatibility with strong segmentation backbones such as MedSAM. Importantly, *DeferredSeg* selectively routes only a small subset of pixels to expert branches during inference, as quantified by the pixel-level deferral ratio. In an intended prospective workflow, experts could focus on the difficult regions assigned by the deferral function rather than inspecting the entire image, although the actual efficiency of this workflow remains to be evaluated.

Several aspects remain worth further investigation. First, although *DeferredSeg* reduces the proportion of pixels assigned to expert branches during inference, its training stage still relies on previously annotated expert data. Therefore, the availability and annotation cost of high-quality expert labels remain practical factors to consider. Second, the effectiveness of *DeferredSeg* may be influenced by the reliability, regional expertise, and complementarity of the available experts. Third, our current real-expert evaluation uses pre-existing clinician annotations from Chaksu, QUBIQ, and CURVAS in an offline setting. Broader prospective studies across more organs, imaging modalities, institutions, and clinical tasks are still needed to assess clinical generality and interaction efficiency. Finally, practical deployment will require careful integra-

tion of deferred-region expert annotation into clinical workflows, including efficient interaction interfaces, manageable review time, compatibility with existing clinical systems.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI and AI-assisted technologies for grammar, formatting, and spelling checks. The authors reviewed and edited the manuscript and take full responsibility for its content.

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Appendix A. Interactive Expert Annotation Interface

To demonstrate the intended inference-time Human–AI collaboration workflow, we implement a proof-of-concept Streamlit interface that allows experts to provide local corrections within regions deferred by *DeferredSeg*. The interface demonstrates the technical workflow of expert-region assignment, local correction, and final System fusion. However, its usability, annotation time, interaction cost, and clinical efficiency remain to be evaluated through prospective clinician studies. The interface is organised into three stages (see Fig. A.5):

- **Step 1 – Model inference & expert assignment.** The user uploads a slice (or selects one from the built-in dataset browser). *DeferredSeg* runs the MedSAM-based segmentor and routing function to obtain the base prediction and pixel-wise routing scores, and automatically partitions the image into J expert regions plus a model-only region.

- **Step 2 – Expert interactive annotation.** For each expert region, the interface creates a dedicated canvas that only highlights pixels assigned to that expert. The human expert draws corrections *within their own assigned region* using freehand tools, providing refined labels exactly where their expertise is requested, while non-assigned areas remain untouched in that canvas.

- **Step 3 – Fusion & comparison.** The corrected expert masks are fused with the model branch via the same deferral rule as in the main framework, producing a final segmentation mask. The interface visualises the fused result and, when available, overlays the ground-truth mask for qualitative comparison.

A screen-capture video demonstrating the full workflow of this interface (including loading data, assigning expert regions, annotating, and fusing the final result) is provided in the supplementary material.

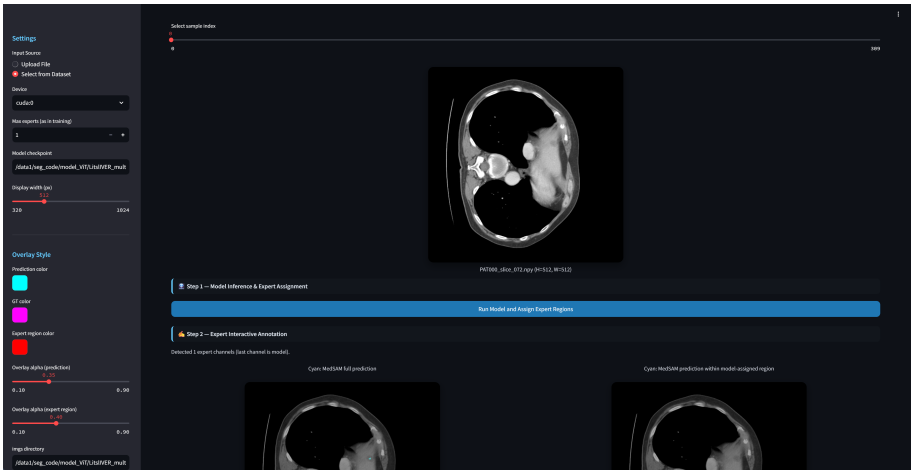


Figure A.5: *DeferredSeg* interactive expert-annotation interface. The page shows (top) model prediction and routing results, (middle) per-expert canvases for interactive corrections, and (bottom) the fused segmentation and ground-truth comparison.

Appendix B. Dataset Details

PROMISE12 The PROMISE12 challenge provides 50 T2-weighted MRI volumes for training and 30 hidden test cases, each annotated with a whole-prostate mask[1]. We resample all volumes to 1 mm^3 isotropic voxel spacing. This dataset targets single-organ segmentation in MRI and serves as a testbed for assessing deferral sensitivity near ambiguous prostate boundaries.

LiTS The LiTS challenge offers 201 abdominal CT scans, with 131 publicly labeled volumes containing both liver and tumour masks, and 70 held-out cases for blind testing[2]. We clip CT intensities to $[-250, 250]$ Hounsfield units (HU) and linearly scale the clipped values to the $[0, 1]$ range. This dataset tests the model's ability to balance large-organ and small-lesion segmentation under significant scale disparity.

AMOS22 AMOS22 contains 500 CT and 100 MRI scans, each annotated for 15 abdominal organs[3]. We follow the official split, resample all volumes to 1 mm

[1] `https://promise12.grand-challenge.org/`
[2] `https://kaiko-ai.github.io/eva/main/datasets/lits/`
[3] `https://amos22.grand-challenge.org/`

isotropic spacing, and clip CT intensities to $[-250, 250]$ HU, and normalise MRI volumes via whole-volume min–max scaling. This multi-organ dataset provides a comprehensive setting for evaluating scalability and expert routing in complex anatomical contexts.

Appendix C. Implementation Details

Appendix C.0.1. Training

The MedSAM encoder and prompt modules are frozen, and only the reconfigured decoder, routing branches, and deferral predictor are trained using AdamW with a learning rate of 1×10^{-4}, StepLR scheduling, an effective batch size of 8, early stopping, and mixed precision. For all trainable methods reported with mean and standard deviation, we repeat the experiments over three random seeds and report the mean and standard deviation.

Appendix D. Complete Quantitative Results

As a complement to Tab. 3 in the main text, Tab. D.10 reports the complete quantitative results across multiple datasets, organs, and evaluation metrics, including standard deviations as well as the Expert and Model branches.

Appendix E. Complete Real-Expert Evaluation Results

As a complement to Tab. 4 in the main text, Tab. E.11 reports the complete real-expert evaluation results across Chaksu, QUBIQ, and CURVAS. The table includes standard deviations as well as the System, Expert, and Model branch results of *DeferredSeg*.

Appendix F. Cross-center Validation on PROMISE12

To evaluate the robustness of *DeferredSeg* under center shift, we conduct leave-one-center-out cross-center validation on the 50 labeled PROMISE12 training cases.

Table D.10: Comparison with fully automatic reference baselines and the expert-assisted PatchCorr baseline across multiple datasets, organs, and evaluation metrics (%). Row-wise best values are **bold**.

Dataset	Metric	nnUNet v2 (MICCAI 2024)	CENet (MICCAI 2025)	MedSAM (NC 2024)	MedSAM (Fine-tuned)	PatchCorr (MIDL 2024)	Ours (1 expert) System	Ours (1 expert) Expert	Ours (1 expert) Model	Ours (3 experts) System	Ours (3 experts) Expert	Ours (3 experts) Model
Prostate (PROMISE12)	DSC	90.49 ± 0.12	68.96 ± 0.16	84.16	86.17 ± 1.53	92.24 ± 0.03	**97.06 ± 0.05**	97.88 ± 0.01	82.80 ± 2.75	96.00 ± 0.14	95.09 ± 0.19	87.17 ± 1.99
	Jaccard	82.70 ± 0.20	54.62 ± 0.13	73.01	76.08 ± 2.32	85.98 ± 0.05	**94.38 ± 0.08**	95.85 ± 0.01	79.56 ± 2.68	92.46 ± 0.24	90.80 ± 0.32	86.80 ± 1.91
	Sens.	91.78 ± 0.53	75.20 ± 0.10	76.70	84.97 ± 6.97	94.45 ± 0.04	95.33 ± 0.09	97.01 ± 0.00	83.55 ± 1.37	**96.42 ± 0.04**	95.57 ± 0.04	99.05 ± 0.64
Liver (LiTS)	DSC	96.62 ± 0.05	93.61 ± 0.10	91.64	91.69 ± 0.06	92.86 ± 0.00	**97.96 ± 0.15**	95.38 ± 0.09	95.07 ± 0.14	96.15 ± 0.51	97.17 ± 0.31	93.69 ± 0.98
	Jaccard	93.48 ± 0.10	89.52 ± 0.07	86.43	86.49 ± 0.07	88.83 ± 0.01	**96.30 ± 0.16**	91.34 ± 0.07	94.13 ± 0.13	93.70 ± 0.64	95.16 ± 0.35	91.01 ± 1.04
	Sens.	**98.37 ± 0.11**	93.85 ± 0.14	93.98	94.06 ± 0.04	98.08 ± 0.01	96.83 ± 0.23	92.02 ± 0.01	95.57 ± 0.60	95.47 ± 0.60	97.03 ± 0.03	93.16 ± 0.65
Tumor (LiTS)	DSC	68.04 ± 1.88	75.35 ± 0.52	59.62	59.20 ± 0.26	65.42 ± 0.05	**91.01 ± 4.30**	93.88 ± 3.16	67.34 ± 25.33	81.82 ± 12.90	83.51 ± 9.61	68.52 ± 5.86
	Jaccard	52.72 ± 1.10	64.91 ± 0.36	47.83	47.55 ± 0.14	54.91 ± 0.06	**85.81 ± 6.60**	89.64 ± 5.36	64.03 ± 26.14	74.59 ± 15.04	78.52 ± 6.72	63.42 ± 11.30
	Sens.	84.99 ± 0.52	76.38 ± 0.65	76.18	75.97 ± 0.62	87.75 ± 0.10	**88.99 ± 6.67**	93.81 ± 2.80	73.00 ± 24.24	84.39 ± 16.40	95.65 ± 3.31	80.04 ± 16.58
Spleen	DSC	94.29 ± 0.07	89.62 ± 0.25	89.21	94.57 ± 0.39	93.80 ± 0.00	97.75 ± 0.46	95.52 ± 2.59	97.69 ± 0.06	**98.14 ± 0.49**	95.74 ± 2.47	98.28 ± 0.11
	Jaccard	90.83 ± 0.11	84.79 ± 0.26	81.74	90.77 ± 0.50	89.33 ± 0.00	95.86 ± 0.85	91.64 ± 4.74	96.52 ± 0.06	**96.59 ± 0.93**	92.06 ± 4.57	97.59 ± 0.11
	Sens.	95.21 ± 0.53	89.41 ± 0.28	90.78	95.02 ± 0.37	95.41 ± 0.00	98.06 ± 0.87	95.50 ± 4.95	99.00 ± 0.01	**98.34 ± 0.94**	95.34 ± 4.72	99.61 ± 0.01
Right Kidney	DSC	90.26 ± 0.74	91.34 ± 0.86	89.39	94.83 ± 0.07	93.76 ± 0.00	**98.34 ± 0.68**	95.88 ± 2.60	98.22 ± 0.02	**98.34 ± 0.51**	96.08 ± 2.56	98.30 ± 0.00
	Jaccard	85.99 ± 0.72	86.47 ± 0.93	81.98	90.85 ± 0.08	89.05 ± 0.01	**96.85 ± 1.30**	92.23 ± 4.80	97.55 ± 0.04	96.84 ± 0.97	92.61 ± 4.74	97.66 ± 0.01
	Sens.	85.86 ± 0.51	91.27 ± 1.16	91.70	95.25 ± 0.04	96.46 ± 0.00	**98.69 ± 1.31**	95.51 ± 4.94	99.67 ± 0.01	98.49 ± 0.96	95.42 ± 4.84	99.80 ± 0.02
Left kidney	DSC	93.67 ± 0.29	91.61 ± 1.36	89.63	94.74 ± 0.09	93.42 ± 0.00	98.14 ± 0.63	95.68 ± 2.60	97.91 ± 0.04	**98.16 ± 0.56**	96.12 ± 2.51	98.34 ± 0.01
	Jaccard	88.99 ± 0.37	86.92 ± 1.48	82.33	90.86 ± 0.12	88.63 ± 0.00	96.60 ± 1.18	91.94 ± 4.78	97.13 ± 0.04	**96.63 ± 1.09**	92.72 ± 4.68	97.85 ± 0.01
	Sens.	93.21 ± 1.31	91.39 ± 1.81	93.11	94.67 ± 0.29	97.62 ± 0.00	**98.72 ± 1.23**	95.50 ± 4.95	99.53 ± 0.04	98.25 ± 1.12	95.42 ± 4.84	99.74 ± 0.07
Gallbladder	DSC	76.73 ± 0.21	62.58 ± 3.72	79.60	86.69 ± 0.67	90.82 ± 0.00	**96.57 ± 1.42**	95.19 ± 2.56	90.84 ± 0.25	95.27 ± 1.17	93.75 ± 2.55	92.22 ± 0.67
	Jaccard	68.31 ± 0.28	45.66 ± 3.39	68.58	78.77 ± 0.79	84.62 ± 0.00	**93.88 ± 2.65**	91.29 ± 4.70	90.05 ± 0.22	91.82 ± 2.17	89.19 ± 4.59	91.40 ± 0.70
	Sens.	81.95 ± 0.52	50.48 ± 4.23	79.56	88.46 ± 1.66	92.20 ± 0.01	**97.27 ± 2.90**	95.48 ± 5.00	99.19 ± 0.01	96.62 ± 2.24	95.40 ± 4.78	99.56 ± 0.17
Esophagus	DSC	82.71 ± 0.05	66.44 ± 0.54	65.72	79.67 ± 0.51	90.96 ± 0.01	96.14 ± 1.81	95.62 ± 2.59	89.05 ± 0.31	**96.44 ± 1.78**	95.82 ± 2.48	89.44 ± 0.01
	Jaccard	72.07 ± 0.05	55.24 ± 0.59	51.50	68.37 ± 0.09	84.10 ± 0.01	92.78 ± 3.36	91.76 ± 4.78	87.19 ± 0.30	**93.28 ± 3.33**	92.14 ± 4.60	87.96 ± 0.01
	Sens.	84.43 ± 0.15	65.63 ± 0.73	67.36	78.58 ± 0.31	88.46 ± 0.01	**96.50 ± 3.48**	95.50 ± 4.95	96.98 ± 0.17	96.43 ± 3.46	95.36 ± 4.75	98.25 ± 0.16
Liver (AMOS22)	DSC	97.17 ± 0.05	90.90 ± 0.43	86.78	95.15 ± 0.52	89.29 ± 0.00	97.54 ± 0.49	95.40 ± 2.56	96.93 ± 0.02	**97.87 ± 0.45**	95.65 ± 2.47	97.15 ± 0.07
	Jaccard	94.57 ± 0.09	86.80 ± 0.44	78.67	91.82 ± 0.11	82.85 ± 0.00	95.48 ± 0.93	91.42 ± 4.71	95.40 ± 0.01	**96.07 ± 0.86**	91.85 ± 4.52	95.87 ± 0.06
	Sens.	97.37 ± 0.05	90.64 ± 0.73	91.25	95.44 ± 0.84	95.99 ± 0.00	98.44 ± 0.96	95.50 ± 4.95	98.92 ± 0.03	**98.70 ± 0.87**	96.26 ± 4.70	99.20 ± 0.07
Stomach	DSC	87.27 ± 0.30	77.07 ± 0.60	81.56	90.69 ± 0.41	89.81 ± 0.00	**96.70 ± 0.72**	95.54 ± 2.56	95.97 ± 0.06	96.67 ± 0.57	95.62 ± 2.36	95.73 ± 0.04
	Jaccard	80.14 ± 0.36	70.37 ± 0.50	70.82	84.98 ± 0.28	83.32 ± 0.00	**94.03 ± 1.36**	91.73 ± 4.72	94.27 ± 0.06	93.98 ± 1.08	91.85 ± 4.36	93.73 ± 0.01
	Sens.	89.78 ± 0.43	76.35 ± 0.81	81.04	90.46 ± 0.07	92.84 ± 0.01	**97.62 ± 1.41**	95.51 ± 4.94	99.02 ± 0.11	**97.62 ± 1.15**	95.12 ± 4.49	98.71 ± 0.03
Aorta	DSC	94.85 ± 0.06	86.35 ± 3.42	86.45	94.61 ± 0.87	93.57 ± 0.00	98.00 ± 0.50	95.95 ± 2.57	98.72 ± 0.02	**98.20 ± 0.42**	95.97 ± 2.52	98.72 ± 0.01
	Jaccard	90.40 ± 0.09	80.95 ± 3.51	78.02	90.36 ± 1.02	89.90 ± 0.00	96.34 ± 0.98	92.38 ± 4.78	97.96 ± 0.03	**96.68 ± 0.82**	92.46 ± 4.68	97.91 ± 0.02
	Sens.	93.75 ± 0.22	85.29 ± 4.15	87.59	94.35 ± 0.71	94.69 ± 0.00	98.11 ± 1.02	95.49 ± 4.94	99.33 ± 0.02	**98.47 ± 0.88**	95.40 ± 4.82	99.57 ± 0.03
Inferior vena cava	DSC	87.84 ± 0.03	75.42 ± 1.07	77.59	85.42 ± 0.40	92.37 ± 0.00	**97.02 ± 0.93**	95.96 ± 2.57	97.30 ± 0.05	97.01 ± 0.81	96.04 ± 2.48	97.10 ± 0.07
	Jaccard	78.84 ± 0.07	66.16 ± 1.15	65.58	76.61 ± 0.19	86.43 ± 0.00	**94.39 ± 1.78**	92.38 ± 4.77	95.95 ± 0.04	94.38 ± 1.54	92.50 ± 4.60	95.47 ± 0.06
	Sens.	88.42 ± 0.17	85.20 ± 1.17	75.48	83.88 ± 0.62	89.20 ± 0.00	97.17 ± 1.82	95.51 ± 4.94	99.26 ± 0.02	**97.23 ± 1.59**	95.30 ± 4.72	99.00 ± 0.02
Pancreas	DSC	82.06 ± 0.23	62.46 ± 1.21	69.30	83.90 ± 0.09	80.51 ± 0.00	**95.68 ± 1.62**	95.50 ± 2.57	90.58 ± 0.56	95.39 ± 1.25	95.57 ± 2.43	91.02 ± 0.40
	Jaccard	71.11 ± 0.30	47.18 ± 1.67	56.06	74.11 ± 0.58	71.06 ± 0.00	**92.19 ± 3.01**	91.72 ± 4.74	87.93 ± 0.52	91.67 ± 2.32	91.86 ± 4.50	88.38 ± 0.43
	Sens.	85.44 ± 0.38	56.61 ± 0.35	77.23	86.92 ± 0.53	90.46 ± 0.01	**96.54 ± 3.17**	95.51 ± 4.95	96.42 ± 0.01	95.82 ± 2.41	95.26 ± 4.63	97.02 ± 0.04
Duodenum	DSC	74.75 ± 0.10	38.54 ± 9.32	62.14	79.83 ± 0.85	75.44 ± 0.00	94.63 ± 1.76	95.44 ± 2.56	86.81 ± 0.11	**94.86 ± 1.56**	95.64 ± 2.38	88.07 ± 0.09
	Jaccard	62.07 ± 0.13	25.74 ± 7.95	48.34	68.50 ± 0.32	65.21 ± 0.01	90.38 ± 3.21	91.47 ± 4.71	83.03 ± 0.08	**90.72 ± 2.84**	91.81 ± 4.40	84.27 ± 0.08
	Sens.	74.59 ± 0.15	30.89 ± 8.65	69.37	81.53 ± 0.62	82.02 ± 0.02	**95.40 ± 3.37**	95.49 ± 4.96	93.16 ± 0.09	95.38 ± 3.03	95.18 ± 4.55	93.58 ± 0.01
Bladder	DSC	87.11 ± 0.54	75.38 ± 13.87	81.64	89.17 ± 0.57	89.50 ± 0.00	**95.99 ± 1.18**	95.65 ± 2.58	92.98 ± 0.11	**95.99 ± 1.10**	95.46 ± 2.48	92.45 ± 0.04
	Jaccard	80.74 ± 0.41	61.78 ± 16.36	71.80	82.72 ± 0.89	83.42 ± 0.00	93.27 ± 2.15	91.98 ± 4.76	91.42 ± 0.12	**93.29 ± 2.02**	91.75 ± 4.54	90.90 ± 0.03
	Sens.	94.39 ± 0.35	69.68 ± 15.57	81.94	92.45 ± 0.26	93.21 ± 0.01	**97.68 ± 2.26**	95.50 ± 4.95	98.61 ± 0.31	97.62 ± 2.18	95.32 ± 4.78	98.62 ± 0.07
Prostate /Uterus	DSC	85.40 ± 0.23	41.64 ± 14.05	81.11	88.73 ± 0.15	93.16 ± 0.01	97.09 ± 0.98	95.69 ± 1.82	97.74 ± 0.03	**97.19 ± 0.90**	95.72 ± 2.50	97.32 ± 0.15
	Jaccard	75.83 ± 0.31	29.57 ± 12.14	69.65	80.77 ± 0.61	87.77 ± 0.02	94.58 ± 1.83	91.90 ± 3.35	96.89 ± 0.09	**94.78 ± 1.72**	92.02 ± 4.62	96.52 ± 0.15
	Sens.	88.46 ± 0.16	36.72 ± 15.53	76.65	91.44 ± 0.08	92.42 ± 0.01	97.21 ± 1.85	95.50 ± 3.50	99.42 ± 0.06	**97.26 ± 1.80**	95.42 ± 4.82	99.39 ± 0.06

Table E.11: Complete real-expert evaluation across Chaksu, QUBIQ, and CURVAS (mean ± standard deviation, %). Fully automatic methods are included as model-only references, while PatchCorr is included as an expert-assisted baseline. Row-wise best full-image results are in **bold**. Expert- and Model-branch results are shown in gray.

Dataset	Metric	CENet MICCAI 2025	MedSAM NC 2024	MedSAM Fine-tuned	PatchCorr MIDL 2024	Ours (1 expert) System	Ours (1 expert) Expert	Ours (1 expert) Model	Ours (3 experts) System	Ours (3 experts) Expert	Ours (3 experts) Model
Chaksu	DSC	97.20 ± 0.06	94.97	96.76 ± 0.08	90.37 ± 0.07	**97.36 ± 0.05**	97.37 ± 0.04	96.50 ± 0.10	97.17 ± 0.04	97.27 ± 0.05	96.98 ± 0.15
	Jaccard	94.64 ± 0.08	90.68	93.89 ± 0.13	82.78 ± 0.04	**94.87 ± 0.01**	94.87 ± 0.03	93.56 ± 0.08	94.60 ± 0.07	94.74 ± 0.11	94.49 ± 0.09
	Sens.	96.67 ± 0.15	91.51	96.69 ± 0.11	84.11 ± 0.12	96.30 ± 0.04	96.50 ± 0.04	94.55 ± 0.05	**96.74 ± 0.25**	95.72 ± 0.10	96.84 ± 0.29
Brain Growth (QUBIQ)	DSC	86.76 ± 0.05	66.98	83.15 ± 0.50	82.10 ± 0.47	86.68 ± 0.93	92.32 ± 0.86	84.87 ± 0.29	**89.33 ± 1.34**	87.45 ± 2.06	89.50 ± 1.79
	Jaccard	79.10 ± 0.08	50.57	71.28 ± 0.67	72.37 ± 0.75	78.52 ± 1.37	86.11 ± 1.51	73.96 ± 0.38	**81.51 ± 1.89**	78.18 ± 3.34	81.05 ± 2.45
	Sens.	86.04 ± 0.23	80.86	85.03 ± 0.46	84.38 ± 0.40	86.49 ± 1.51	92.23 ± 1.03	86.72 ± 1.12	**88.20 ± 2.95**	86.14 ± 3.41	90.46 ± 3.60
Prostate (QUBIQ)	DSC	94.23 ± 0.40	93.39	94.77 ± 0.20	95.62 ± 0.04	**96.73 ± 0.28**	83.73 ± 2.07	94.87 ± 0.53	95.71 ± 0.15	97.37 ± 0.39	95.27 ± 1.44
	Jaccard	89.36 ± 0.66	89.24	90.16 ± 0.37	91.68 ± 0.09	**94.56 ± 0.49**	77.77 ± 3.81	91.59 ± 0.91	92.17 ± 0.28	94.96 ± 0.74	91.54 ± 2.52
	Sens.	**96.72 ± 0.46**	92.16	96.20 ± 0.32	96.29 ± 0.17	93.69 ± 1.32	87.35 ± 1.24	95.57 ± 1.83	92.76 ± 0.73	97.80 ± 0.23	93.12 ± 3.21
Kidney (CURVAS)	DSC	93.17 ± 0.43	67.94	90.70 ± 0.31	91.32 ± 0.17	94.48 ± 0.20	99.64 ± 0.26	89.83 ± 0.21	**95.80 ± 0.31**	99.37 ± 0.30	89.58 ± 0.24
	Jaccard	88.78 ± 0.60	54.93	84.41 ± 0.37	86.75 ± 0.21	90.00 ± 0.20	99.52 ± 0.29	83.36 ± 0.19	**91.98 ± 0.31**	99.12 ± 0.37	83.09 ± 0.22
	Sens.	92.45 ± 0.40	88.90	91.83 ± 0.66	89.49 ± 0.18	92.67 ± 0.23	99.64 ± 0.21	90.09 ± 0.24	**93.97 ± 0.54**	99.29 ± 0.32	89.89 ± 0.51
Liver (CURVAS)	DSC	93.84 ± 0.16	92.51	93.14 ± 0.29	92.39 ± 0.14	93.55 ± 0.09	99.52 ± 0.07	93.28 ± 0.20	**95.80 ± 0.08**	99.47 ± 0.04	93.13 ± 0.10
	Jaccard	88.43 ± 0.28	87.74	88.39 ± 0.35	87.24 ± 0.18	89.93 ± 0.13	99.40 ± 0.09	88.56 ± 0.23	**91.94 ± 0.02**	99.15 ± 0.09	88.39 ± 0.08
	Sens.	93.65 ± 0.19	92.68	94.17 ± 0.18	92.16 ± 0.09	**94.88 ± 0.24**	99.71 ± 0.08	93.74 ± 0.36	94.04 ± 0.24	99.46 ± 0.11	93.61 ± 0.30

Table F.12: Leave-one-center-out cross-center validation on PROMISE12. Values are reported as percentages. MedSAM Zero-shot denotes direct inference using the original MedSAM checkpoint. MedSAM-FT denotes supervised fine-tuning on the training centers. The best result for each metric is highlighted in bold.

Held-out site	Cases	MedSAM Zero-shot			MedSAM-FT			Ours-1E			Ours-3E		
		DSC	Jaccard	Sens.	DSC	Jaccard	Sens.	DSC	Jaccard	Sens.	DSC	Jaccard	Sens.
Site A	15	84.88	74.31	**89.45**	86.10	76.16	81.23	87.16	77.24	82.46	**91.48**	**84.95**	85.30
Site B	15	84.58	74.04	86.73	81.39	69.33	71.81	84.15	72.64	76.41	**93.61**	**88.19**	**89.99**
Site C	15	83.46	72.23	**88.00**	84.81	74.00	77.87	86.11	75.61	80.11	**92.33**	**86.18**	85.79
Site D	5	80.89	68.57	**89.24**	86.29	76.23	80.47	88.80	79.86	79.63	**93.05**	**87.26**	87.83
Center average	–	83.45	72.29	**88.35**	84.65	73.93	77.84	86.56	76.34	79.65	**92.62**	**86.65**	87.23
Case-weighted average	50	83.97	73.03	**88.18**	84.32	73.47	77.32	86.11	75.63	79.66	**92.53**	**86.52**	87.11

Table F.13: Deferral ratio in leave-one-center-out cross-center validation on PROMISE12. Values are reported as percentages. Deferred denotes the percentage of pixels routed to expert branches.

Held-out site	Ours-1E		Ours-3E				
	Deferred	Model	Deferred	E1	E2	E3	Model
Site A	2.53	97.47	6.06	0.62	4.31	1.13	93.94
Site B	2.17	97.83	6.25	0.44	4.90	0.91	93.75
Site C	1.95	98.05	5.34	0.52	3.84	0.99	94.66
Site D	1.71	98.29	4.34	0.37	3.09	0.88	95.66
Center average	2.09	97.91	5.50	0.49	4.04	0.98	94.50
Case-weighted average	2.17	97.83	5.73	0.51	4.22	1.00	94.27

PROMISE12 was originally collected from four acquisition centers with differences in scanner manufacturer, field strength, and imaging protocol. We denote the four site-specific subsets as Site A–D. In each fold, one site is held out for testing and the remaining sites are used for training. We compare MedSAM Zero-shot, supervised fine-tuned MedSAM, and *DeferredSeg* with one or three expert branches.

As shown in Tab. F.12, the three-expert setting achieves the best DSC and Jaccard on all held-out sites. Compared with MedSAM Zero-shot and MedSAM-FT, *DeferredSeg* consistently improves overlap-based performance, with the three-expert setting showing the largest gain in the case-weighted average. These results indicate that pixel-wise deferral can improve segmentation robustness under cross-center distribution shifts.

Tab. F.13 further reports the expert routing allocation. The three-expert setting defers only a small fraction of pixels to expert branches, ranging from 4.34% to 6.25% across held-out sites, with a case-weighted average deferral ratio of 5.73%. This means that most pixels are still handled by the model branch.

Appendix G. Additional Qualitative Visualization

To further demonstrate the generalizability of *DeferredSeg*, we provide additional qualitative visualization results on multiple datasets and imaging modalities in Fig. G.6. The supplementary examples cover multiple AMOS22 organs from abdominal CT/MRI scans, including spleen, right kidney, gallbladder, liver, and stomach, liver tumor CT cases from LiTS, and a retinal fundus example from Chaksu. Compared with the automated model branch, the System prediction of *DeferredSeg* better handles ambiguous boundaries and difficult foreground regions by learning to identify difficult pixels and defer them to more reliable expert branches. These examples complement Fig. 3(a) and provide broader qualitative evidence for the effectiveness and generalizability of the proposed Human–AI collaborative segmentation framework.

Appendix H. Detailed Statistical Significance Analysis

Following the statistical testing protocol described in the main text, we report the complete dataset-level results of the one-sided paired Wilcoxon signed-rank tests on case-level DSC scores.

For the MedSAM-based experiments, the one- and three-expert configurations of *DeferredSeg* are separately compared with fine-tuned MedSAM (MedSAM-FT) and PatchCorr on the same test cases. For the CENet-based experiments, CENet with *DeferredSeg* is compared with the corresponding fine-tuned CENet baseline under the same backbone.

For each test case i, W/T/L denotes whether the case-level DSC of *DeferredSeg* is higher than, equal to, or lower than that of baseline b, respectively:

$$\mathrm{W} : \mathrm{DSC}_{DeferredSeg,i} > \mathrm{DSC}_{b,i}, \qquad \mathrm{T} : \mathrm{DSC}_{DeferredSeg,i} = \mathrm{DSC}_{b,i}, \qquad \mathrm{L} : \mathrm{DSC}_{DeferredSeg,i} < \mathrm{DSC}_{b,i}.$$

Thus, W/T/L reports the numbers of winning, tied, and losing test cases. The null and

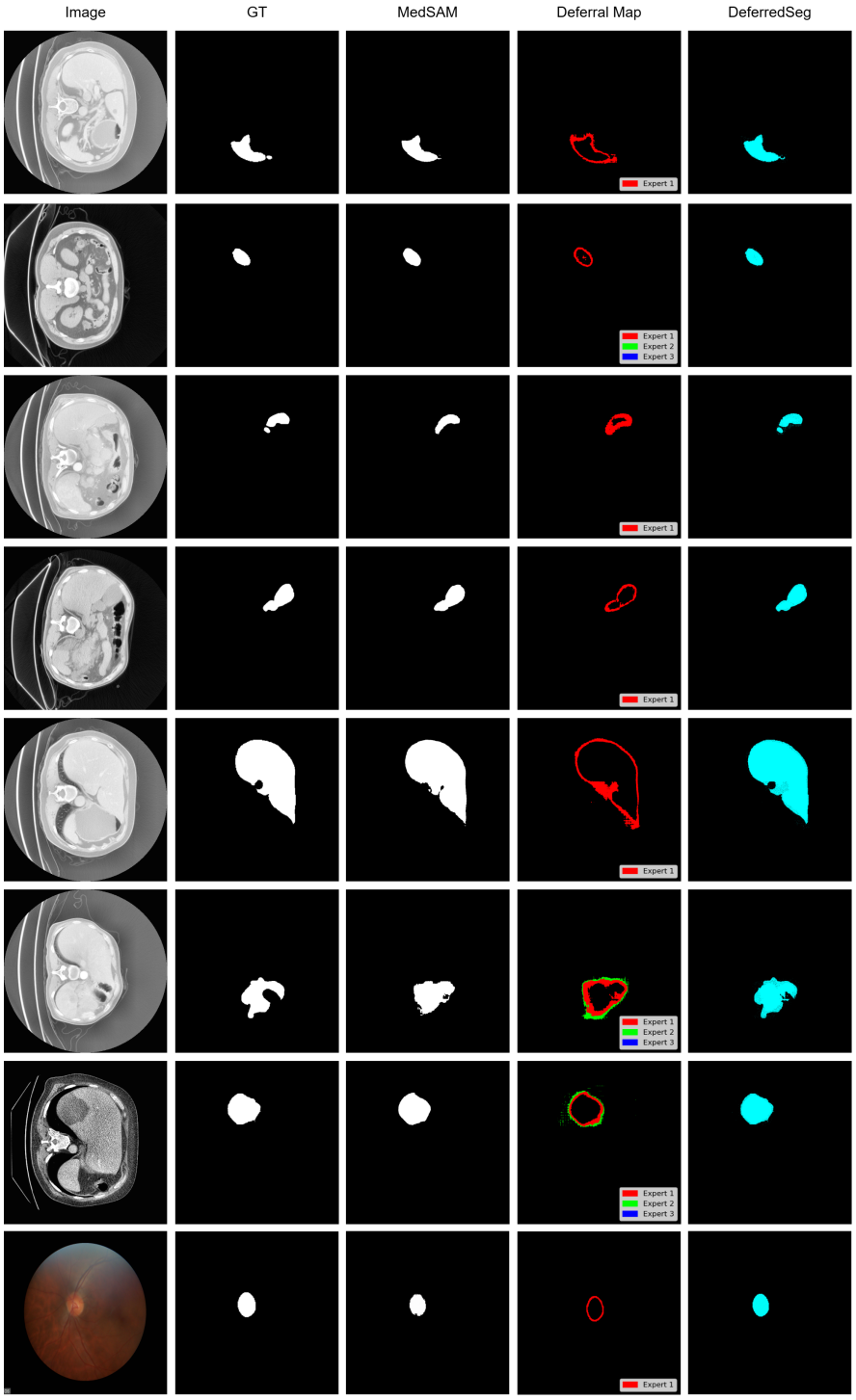


Figure G.6: Additional qualitative visualization results on different datasets, anatomical structures, and imaging modalities.

alternative hypotheses are

$$H_0 : \Delta_b \leq 0, \qquad H_1 : \Delta_b > 0,$$

where Δ_b denotes the location shift of the paired DSC differences relative to baseline b. Therefore, H_1 indicates that *DeferredSeg* achieves higher case-level DSC than the corresponding baseline. Statistical significance is determined at $p < 0.05$, and n.s. denotes a non-significant result.

Table H.14: Detailed statistical significance analysis of the MedSAM-based one- and three-expert configurations using one-sided paired Wilcoxon signed-rank tests on case-level DSC. Each configuration is compared with fine-tuned MedSAM (MedSAM-FT) and PatchCorr on the same test cases. W/T/L denotes the numbers of test cases in which *DeferredSeg* obtains higher/equal/lower case-level DSC than the corresponding baseline, respectively. n.s. denotes $p \geq 0.05$.

Dataset	Cases	One expert				Three experts			
		vs. MedSAM-FT		vs. PatchCorr		vs. MedSAM-FT		vs. PatchCorr	
		W/T/L	p	W/T/L	p	W/T/L	p	W/T/L	p
PROMISE12	30	29/0/1	< .001	27/0/3	< .001	29/0/1	< .001	27/0/3	< .001
LiTS-Liver	28	27/0/1	< .001	25/0/3	< .001	15/0/13	n.s.	21/0/7	< .01
LiTS-Tumor	27	26/0/1	< .001	25/0/2	< .001	25/0/2	< .001	24/0/3	< .001
AMOS22-Spleen	120	96/0/24	< .001	94/0/26	< .001	99/0/21	< .001	98/0/22	< .001
AMOS22-Right Kidney	118	104/0/14	< .001	101/0/17	< .001	103/0/15	< .001	99/0/19	< .001
AMOS22-Left Kidney	120	108/0/12	< .001	105/0/15	< .001	106/0/14	< .001	104/0/16	< .001
AMOS22-Gallbladder	112	103/0/9	< .001	98/0/14	< .001	106/0/6	< .001	101/0/11	< .001
AMOS22-Esophagus	120	118/0/2	< .001	112/0/8	< .001	119/0/1	< .001	115/0/5	< .001
AMOS22-Liver	120	63/0/57	n.s.	112/0/8	< .001	98/0/22	< .001	116/0/4	< .001
AMOS22-Stomach	119	103/0/16	< .001	101/0/18	< .001	110/0/9	< .001	108/0/11	< .001
AMOS22-Aorta	120	114/0/6	< .001	111/0/9	< .001	117/0/3	< .001	115/0/5	< .001
AMOS22-Inferior Vena Cava	120	119/0/1	< .001	115/0/5	< .001	120/0/0	< .001	116/0/4	< .001
AMOS22-Pancreas	120	117/0/3	< .001	115/0/5	< .001	119/0/1	< .001	117/0/3	< .001
AMOS22-Duodenum	119	115/0/4	< .001	113/0/6	< .001	112/0/7	< .001	110/0/9	< .001
AMOS22-Bladder	99	91/0/8	< .001	89/0/10	< .001	90/0/9	< .001	87/0/12	< .001
AMOS22-Prostate/Uterus	97	94/0/3	< .001	91/0/6	< .001	93/0/4	< .001	90/0/7	< .001

As shown in Tab. H.14, the one- and three-expert configurations each achieve statistically significant improvements over MedSAM-FT in 15 of the 16 comparisons. The two non-significant cases are AMOS22-Liver under the one-expert setting and LiTS-Liver under the three-expert setting. Compared with PatchCorr, both configurations achieve statistically significant improvements in all 16 dataset-level comparisons.

As shown in Tab. H.15, adding *DeferredSeg* to the CENet backbone produces statistically significant improvements over CENet-FT on all six evaluated segmentation

Table H.15: Statistical significance analysis of the CENet-based instantiation of *DeferredSeg*. One-sided paired Wilcoxon signed-rank tests are performed on case-level DSC between CENet with *DeferredSeg* and the corresponding fine-tuned CENet baseline under the same backbone. W/T/L denotes the numbers of test cases in which CENet with *DeferredSeg* obtains higher/equal/lower case-level DSC than CENet-FT, respectively. n.s. denotes $p \geq 0.05$.

Dataset	Cases	W/T/L	p
PROMISE12	30	29/0/1	$< .001$
LiTS-Liver	28	20/0/8	$< .05$
LiTS-Tumor	27	22/0/5	$< .01$
AMOS22-Spleen	120	98/0/22	$< .001$
Right Kidney	118	101/0/17	$< .001$
Chaksu	273	206/0/67	$< .05$

tasks.

Appendix I. Detailed Ablation Results

As a complement to Tab. 8 in the main text, Tab. I.16 reports the complete ablation results, including the System, Expert, and Model branches under different loss configurations. These detailed results further verify that the proposed components are mutually complementary and indispensable.

Table I.16: Analysis of each loss component through ablation. These components are mutually complementary and indispensable.

J	Loss	DSC			Jaccard			Sens.		
		System	Expert	Model	System	Expert	Model	System	Expert	Model
1	DC	96.86±0.92	95.67±1.48	94.55±0.89	94.00±1.73	91.78±2.72	93.67±1.01	96.80±1.89	95.32±2.90	96.49±1.71
	+ SC	**97.17±0.63**	**96.09±0.88**	**96.08±0.52**	**94.57±1.17**	**92.55±1.62**	**95.14±0.54**	**97.29±1.24**	**96.01±1.76**	**97.41±1.32**
3	DC	95.39±1.64	93.49±2.63	94.35±1.41	91.51±2.95	88.23±4.57	93.58±1.46	97.13±2.23	95.76±3.51	**98.42±1.84**
	+ SC	95.79±1.71	93.98±2.58	94.16±0.76	92.14±3.07	88.98±4.51	93.38±0.70	97.18±2.31	95.79±3.55	97.39±3.01
	+ LB	**96.59±1.46**	95.06±2.49	95.72±0.84	93.52±2.69	90.78±4.45	94.70±0.89	97.16±2.19	95.78±3.52	97.94±1.55
	+ SC, LB	96.58±1.04	**95.08±1.78**	**96.19±0.55**	**93.56±1.87**	**90.85±3.08**	**95.25±0.56**	**98.34±0.47**	**97.61±0.68**	98.39±1.03

Appendix J. Full Results of Coherence-Balance grid search

Table J.17 reports the complete numerical results for all branches (System / Expert / Model) and metrics (Jaccard / DSC / Sensitivity) across all λ_1, λ_2 combinations.

Full Heatmaps: For completeness, Fig. J.7 shows the *nine* heatmaps spanning all branches (System / Expert / Model) and metrics (Jaccard / DSC / Sensitivity).

Table J.17: Effect of varying coherence (λ_1) and balance (λ_2) weights on System / Expert / Model performance (%).

Group	λ_1	λ_2	**System**			**Expert**			**Model**		
			Jaccard	DSC	Sens.	Jaccard	DSC	Sens.	Jaccard	DSC	Sens.
A	0.10	0.10	91.26	95.35	94.74	87.45	93.21	91.99	92.71	93.32	98.55
B	0.10	1.00	91.12	95.25	94.73	87.51	93.25	92.00	93.80	94.43	**99.90**
C	0.10	5.00	91.04	95.11	94.76	87.42	93.09	91.99	93.66	94.47	95.20
D	0.10	10.00	91.36	95.43	94.67	87.89	93.50	92.02	93.12	93.95	99.23
E	1.00	0.10	91.37	95.44	94.66	87.96	93.54	91.99	93.02	93.84	99.39
F	1.00	1.00	90.74	95.02	94.66	87.08	92.97	91.97	93.85	94.59	99.45
G	1.00	5.00	**91.90**	**95.63**	**96.13**	**88.48**	**93.73**	**93.98**	93.20	94.32	94.86
H	1.00	10.00	91.03	95.21	94.70	87.27	93.09	92.01	93.47	94.21	98.91
I	5.00	0.10	90.28	94.61	94.77	86.43	92.41	92.00	93.70	94.44	99.59
J	5.00	1.00	91.63	95.58	94.83	87.90	93.50	92.01	92.57	93.30	99.09
K	5.00	5.00	91.12	95.22	94.66	87.71	93.33	91.99	93.71	94.51	95.02
L	5.00	10.00	91.64	95.59	94.60	88.22	93.69	92.01	91.76	92.42	99.24
M	10.00	0.10	91.30	95.36	94.64	87.88	93.45	92.01	92.14	92.94	98.18
N	10.00	1.00	91.13	95.28	94.62	87.65	93.35	91.98	92.20	93.19	98.39
O	10.00	5.00	91.22	95.34	94.68	87.64	93.32	92.01	**94.06**	**94.88**	99.49
P	10.00	10.00	90.53	94.87	94.68	86.91	92.84	92.00	90.66	91.32	99.55

Appendix J.1. Additional Hyperparameter Analysis on More Datasets

To further examine the robustness of the hyperparameter choice, we conduct an additional sensitivity analysis on LiTS-Liver under the three-expert setting by varying the coherence and balance weights (λ_1, λ_2). As shown in Tab. J.18, the default setting $(\lambda_1, \lambda_2) = (1, 5)$, selected from the PROMISE12 grid search, achieves the best overlap-based performance among the tested configurations, with the highest DSC and Jaccard.

Table J.18: Effect of varying coherence (λ_1) and balance (λ_2) weights on System performance on LiTS-Liver (%).

Group	λ_1	λ_2	**System**		
			DSC	Jaccard	Sens.
A	0.10	0.10	94.15	88.95	92.97
B	0.10	5.00	95.82	91.98	94.10
C	1.00	1.00	95.95	92.22	94.77
D	1.00	5.00	**96.15**	**93.70**	**95.47**
E	1.00	10.00	96.01	92.33	94.88
F	5.00	1.00	94.66	89.86	93.48
G	5.00	5.00	93.86	88.43	92.68
H	10.00	10.00	91.10	83.65	89.92

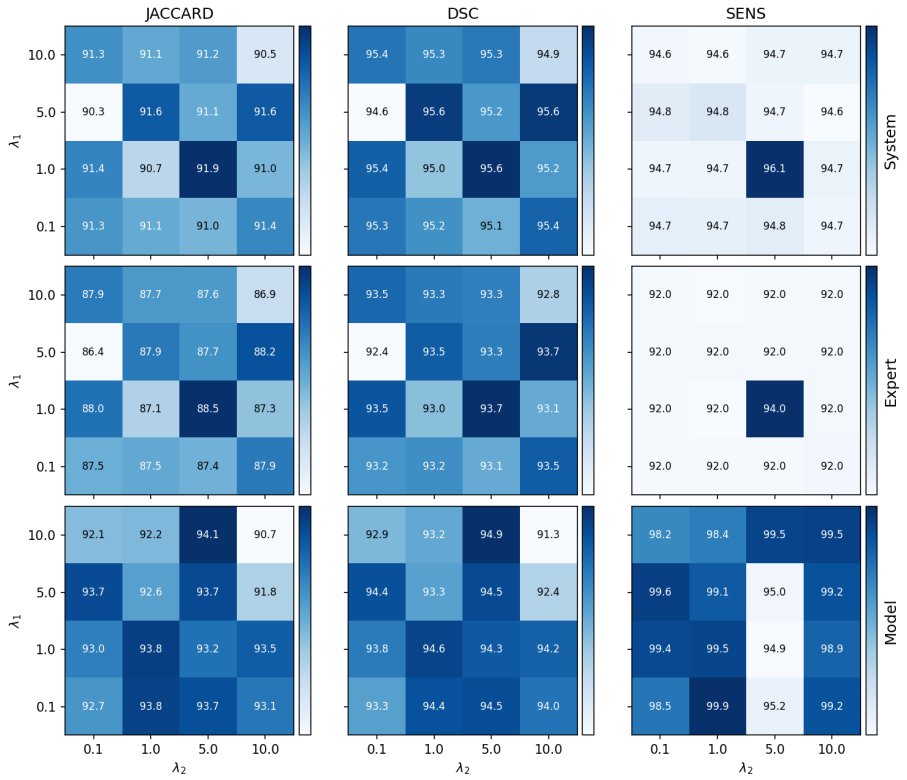


Figure J.7: Complete set of heatmaps for all branches and metrics.

Appendix K. Expert Scalability Analysis

For the scalability experiments, each expert set is drawn from a fixed pool of five synthetic experts (E1-E5), respectively (see Tab. K.19), while keeping the overall FG/BG/BD accuracy approximately constant across groups. Specifically:

- 1 expert uses **E1**.
- 2 experts use **E2, E3**.
- 3 experts use **E1, E2, E3**.
- 4 experts use **E2, E3, E4, E5**.
- 5 experts use **E1, E2, E3, E4, E5**.

The complete System/Expert/Model scores are provided in Tab. K.20

Table K.19: Settings of scalability-specific synthetic expert pool.

Expert	FG	BG	BD
E1	92	98	97
E2	91	99	96
E3	93	97	98
E4	90	97	99
E5	94	99	99

Table K.20: Performance on PROMISE12 under different numbers of experts (%). Single expert yields peak overlap (DSC/Jaccard), whereas multiple experts raise Sensitivity. (mean ± std; row-wise best in **bold**, second best in *italics*)

Metric	Predictor	J = 1	J = 2	J = 3	J = 4	J = 5
DSC	System	**97.06 ± 0.05**	94.48 ± 0.49	*96.00 ± 0.14*	94.43 ± 0.52	95.45 ± 0.31
	Expert	**97.88 ± 0.01**	92.08 ± 0.72	*95.09 ± 0.19*	93.05 ± 0.57	93.48 ± 0.27
	Model	82.80 ± 2.75	**92.14 ± 2.60**	87.17 ± 1.99	82.44 ± 0.61	*91.45 ± 1.84*
Jaccard	System	**94.38 ± 0.08**	89.65 ± 0.83	*92.46 ± 0.24*	89.63 ± 0.89	91.44 ± 0.64
	Expert	**95.85 ± 0.01**	85.42 ± 1.20	*90.80 ± 0.32*	87.16 ± 0.96	87.94 ± 0.94
	Model	79.56 ± 2.68	**91.05 ± 2.30**	86.80 ± 1.91	82.24 ± 0.61	*90.71 ± 1.62*
Sens.	System	95.33 ± 0.09	94.08 ± 0.13	*96.42 ± 0.04*	94.76 ± 0.06	**96.64 ± 0.11**
	Expert	**97.01 ± 0.00**	91.07 ± 0.01	*95.57 ± 0.04*	93.29 ± 0.08	95.00 ± 0.06
	Model	83.55 ± 1.37	*99.21 ± 0.43*	99.05 ± 0.64	**99.75 ± 0.25**	91.95 ± 0.19

Appendix L. Complementary Expert Settings and Detailed Results

To analyze the impact of complementary expert configurations, we define seven synthetic experts with distinct accuracy profiles on FG, BG, and BD regions (Tab. L.21). By selecting different subsets of these experts, we emulate varying levels of complementarity and specialization.

Table L.22 presents the detailed performance of System / Expert / Model branches as the number of complementary experts increases. A single complementary expert (J=1) already achieves high System performance. Adding three or more complementary experts (J=3-5) further improves System Jaccard and DSC, with the strongest results obtained at J=5 (Tab. L.22).

Table L.21: Accuracy profiles of complementary experts.

Expert Type	**FG**	**BG**	**BD**
(E1) FG strong	0.97	0.90	0.98
(E2) BG strong	0.88	0.99	0.96
(E3) BD strong	0.88	0.90	1.00
(E4) FG+BG strong	0.97	0.99	0.98
(E5) FG+BD strong	0.97	0.90	1.00
(E6) BG+BD strong	0.88	0.99	1.00
(E7) FG+BG+BD strong	0.97	0.99	1.00

Table L.22: System / Expert / Model performance with increasing number of complementary experts (mean ± std, %).

J	Experts	System			Expert			Model		
		Jaccard	DSC	Sens.	Jaccard	DSC	Sens.	Jaccard	DSC	Sens.
1	E1	91.04 ± 0.16	95.24 ± 0.10	95.54 ± 0.06	91.00 ± 0.19	95.21 ± 0.10	97.01 ± 0.02	80.81 ± 2.70	83.94 ± 2.84	85.12 ± 1.20
2	E1, E6	89.03 ± 0.72	94.10 ± 0.42	93.58 ± 0.81	91.91 ± 0.09	95.73 ± 0.05	**97.04 ± 0.00**	84.07 ± 5.16	**89.49 ± 5.01**	91.85 ± 2.05
3	E1, E2, E3	92.45 ± 0.21	95.99 ± 0.13	96.38 ± 0.05	90.73 ± 0.26	95.05 ± 0.15	95.51 ± 0.01	**85.21 ± 2.68**	85.59 ± 2.66	98.76 ± 0.97
4	E1, E2, E3, E4	94.06 ± 0.46	96.83 ± 0.18	97.43 ± 0.04	92.02 ± 0.18	95.50 ± 0.12	96.94 ± 0.01	74.45 ± 3.00	74.62 ± 2.98	**99.50 ± 0.44**
5	E1, E2, E3, E4, E5	**94.40 ± 0.45**	**96.88 ± 0.26**	**97.46 ± 0.07**	**93.13 ± 0.36**	**96.39 ± 0.21**	96.92 ± 1.42	78.31 ± 1.58	85.84 ± 3.13	98.48 ± 0.03

Appendix M. Additional Computational Cost Analysis

We further evaluate the computational cost of *DeferredSeg* with one and three experts against the base MedSAM under the same hardware, precision, batch-size, and input-resolution settings. As shown in Tab. M.23, the one-expert configuration increases the parameter count and FLOPs by only 0.63% and 0.24%, respectively, while increasing the average inference time by 4.33%. The three-expert configuration increases the parameter count and FLOPs by 0.94% and 0.26%, respectively, with a 5.94% increase in inference time. These results show that *DeferredSeg* introduces only a small increase in model complexity and computational cost compared with the base MedSAM, while providing the additional deferral capability.

Table M.23: Additional computational-cost comparison with the base MedSAM. FLOPs and inference time are measured with an input resolution of 1024×1024, a batch size of 1, and FP32 precision on the same GPU. Inference time is reported as the mean and standard deviation over five repeated runs.

Method	Params (M)	FLOPs (G)	Time (ms/image)
MedSAM	90.7372	69.2878	16.1161 ± 0.0223
DeferredSeg (1 expert)	91.3125	69.4532	16.8139 ± 0.0157
DeferredSeg (3 experts)	91.5926	69.4676	17.0741 ± 0.0086